



Scientific Committee on Consumer Safety
SCCS

SCIENTIFIC ADVICE
on benzophenone – 2 and benzophenone - 5

(CAS No. 131-55-5 and 6628-37-1,
EC No. 205-028-9 and 613-918-7)



The SCCS adopted this document
during the plenary meeting on 26 June 2025

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

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SCCS members listed below are acknowledged for their valuable contribution to the finalisation of this scientific advice.

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This scientific advice has been subject to a commenting period of eight weeks after its initial publication (from 22 April to 17 June 2025). Comments received during this period were considered by the SCCS. There was no change in the final version except a correction in the background information of the mandate (section 2).

All Declarations of Working Group members are available on the following webpage:
[Register of Commission expert groups and other similar entities \(europa.eu\)](https://ec.europa.eu/sccs/register)

1. ABSTRACT

The SCCS concludes the following:

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

- 1. to identify and justify whether there are specific concerns regarding the use of Benzophenone-2 and Benzophenone-5 in cosmetic products.*

Benzophenone 2 (BP-2)

Having considered all the available data, and the concerns relating to potential endocrine disrupting properties, the SCCS cannot conclude on the safety of Benzophenone-2 because genotoxicity potential cannot be excluded. The data on repeated dose toxicity and reproductive toxicity of Benzophenone-2 are also either limited or not available. The available evidence shows that BP-2 is an endocrine-active substance, due to a clear demonstration of estrogenic activity both in vitro and in vivo.

Benzophenone 5 (BP-5)

Due to the scarcity of information for Benzophenone-5, the SCCS has considered that data from a close analogue BP-4 can be used for read-across for the safety assessment of BP-5. Thus, considering the toxicological data on Benzophenone-4, including the potential endocrine disrupting properties, the SCCS is of the opinion that BP-5 is safe when used as a UV filter up to a maximum concentration of 5% in sunscreen, face and hand cream, lipstick, sunscreen propellant spray and pump spray, when used separately or in combination (based on deterministic aggregated exposure of BP-4).

- 2. to highlight if there is a potential risk for human health from the use of Benzophenone-2 and Benzophenone-5 in cosmetic products.*

The SCCS considers that the use of Benzophenone-2 in cosmetic products may pose a risk to the consumer because of the indications for genotoxicity and endocrine activity. The assessment of safety of BP-2 in cosmetic products also requires appropriate data on repeated dose toxicity and reproductive toxicity.

The SCCS mandate does not address environmental aspects. Therefore, this assessment has not covered the safety of BP-2 and BP-5 for the environment.

Keywords: SCCS, scientific advice, benzophenone – 2, benzophenone-5, Regulation 1223/2009, CAS No. 131-55-5 and 6628-37-1, EC No. 205-028-9 and 613-918-7

Opinion to be cited as: SCCS (Scientific Committee on Consumer Safety), scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products, preliminary version of 27 March 2025, final version of 26 June 2025, SCCS/1679/25.

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SCCS

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Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5)(CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

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2. MANDATE FROM THE EUROPEAN COMMISSION

Background on substances with endocrine disrupting properties

On 7 November 2018, the Commission adopted a review¹ of Regulation (EC) No. 1223/2009 on cosmetic products ('Cosmetics Regulation') regarding substances with endocrine disrupting properties. The review concluded that the Cosmetics Regulation provides the adequate tools to regulate the use of cosmetic substances that present a potential risk for human health, including when displaying ED properties.

The Cosmetics Regulation does not have explicit provisions on EDs. However, it provides a regulatory framework with a view to ensuring a high level of protection of human health. Environmental concerns that substances used in cosmetic products may raise are considered through the application of Regulation (EC) No. 1907/2006 ('REACH Regulation').

In the review, the Commission commits to establishing a priority list of potential EDs not already covered by bans or restrictions in the Cosmetics Regulation for their subsequent safety assessment. A priority list of 28 potential EDs in cosmetics was consolidated in early 2019 based on input provided through a stakeholder consultation. The Commission then organised a public call for data² from 16 May 2019 to 15 October 2019 on 14³ of the 28 substances (to be treated with higher priority) in order to be able to prepare the safety assessment of these substances. Benzophenone-2 (BP-2) and Benzophenone-5 (BP-5) are among the above-mentioned 14 substances for which the call for data took place.

Existing information on Benzophenone-2

The ingredient Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) with the chemical name '2,2',4,4'-tetrahydroxybenzophenone' is currently not regulated under the Cosmetics Regulation 1223/2009. According to CosIng database, Benzophenone-2 is reported to have the following functions: 'UV-filter', 'light stabilizers' and 'fragrance', protecting the cosmetic product from damage (*i.e.*, deterioration effects) caused by light (including UV light) and enhancing the smell of a product and/or perfuming the skin. Benzophenone-2 is suspected of being an endocrine disruptor, while it is also suspected of polluting surface waters and of endangering aquatic organisms.

Existing information on Benzophenone-5

In cosmetic products, the ingredient Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) with the chemical name '2-Hydroxy-4-methoxybenzophenone-5-sodium sulfonate' is currently regulated as a UV-filter (entry 22 of Annex VI) at a maximum concentration of 5 % (as acid). Benzophenone-5 is the sodium salt of Benzophenone-4 that is also regulated under the same entry⁴ and which has been recently re-evaluated by the SCCS in view of its potential endocrine activity. In that Opinion⁵, the SCCS concluded on the safety (under certain conditions) of Benzophenone-4, when used as a UV-filter and light-stabilizer in cosmetic products. In view of its structural similarity to Benzophenone-4 (*i.e.*, being its sodium salt), Benzophenone-5 is suspected of being an endocrine disruptor as well.

¹ <https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018-739-F1-EN-MAIN-PART-1.PDF>

² https://ec.europa.eu/growth/content/call-data-ingredients-potential-endocrine-disrupting-properties-used-cosmetic-products_en

³ Benzophenone-3, kojic acid, 4-methylbenzylidene camphor, propylparaben, triclosan, resorcinol, octocrylene, triclocarban, butylated hydroxytoluene (BHT), benzophenone, homosalate, benzyl salicylate, genistein and daidzein

⁴ the SCCS has recently re-evaluated the safety of Bp-4: https://health.ec.europa.eu/publications/benzophenone-4-cas-no-4065-45-6-ec-no-223-772-2_en

⁵ SCCS (Scientific Committee on Consumer Safety), Opinion on benzophenone – 4 (CAS No. 4065-45-6, EC No. 223-772-2) used in cosmetics products. preliminary version of 14 December 2023, final version of 27 March 2024, SCCS/1660/23

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5)(CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

Given the possible health risks associated with these ingredients, their safe use in cosmetic products should be evaluated by the SCCS in view of technical and scientific progress. The Commission, therefore, requests the SCCS to provide a scientific advice on the safety of Benzophenone-2 and Benzophenone-5 when used in cosmetic products.

Terms of reference

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

1. *to identify and justify whether there are specific concerns regarding the use of Benzophenone-2 and Benzophenone-5 in cosmetic products;*
2. *to highlight if there is a potential risk for human health from the use of Benzophenone-2 and Benzophenone-5 in cosmetic products.*

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

3. SCIENTIFIC ADVICE

Initial dossier: CIR report 2021

3.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

3.1.1 Chemical identity

3.1.1.1 Primary name and/or INCI name

Benzophenone-2 (BP-2)

Benzophenone-5 (BP-5)

3.1.1.2 Chemical names

Benzophenone-2 (BP-2):

2,2',4,4'-tetrahydroxybenzophenone

Other names:

IUPAC name: bis(2,4-dihydroxyphenyl)methanone

(from <https://substances.ineris.fr/substance/131-55-5>, and from <https://echa.europa.eu/fr/substance-information/-/substanceinfo/100.004.573>, consulted on 27 September 2024)

4-(2,4-dihydroxybenzoyl)benzene-1,3-diol

(from <https://echa.europa.eu/fr/substance-information/-/substanceinfo/100.004.573>, consulted on 27 September 2024)

Benzophenone-5 (BP-5):

2-Hydroxy-4-methoxybenzophenone-5-sodium sulfonate

IUPAC Names

Benzenesulfonic acid, 5-benzoyl-4-hydroxy-2-methoxy-, monosodium salt
sodium 5-benzoyl-4-hydroxy-2-methoxybenzene-1-sulfonate

(from <https://echa.europa.eu/fr/substance-information/-/substanceinfo/100.121.934>, IUPAC names, consulted on 27 September 2024)

Other names

Regulatory process names

Benzenesulfonic acid, 5-benzoyl-4-hydroxy-2-methoxy-, monosodium salt

Benzenesulfonic acid, 5-benzoyl-4-hydroxy-2-methoxy-, sodium salt (1:1)

Benzophenone-5

(from <https://echa.europa.eu/fr/substance-information/-/substanceinfo/100.121.934>, regulatory process names, consulted on 27 September 2024)

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

3.1.1.3 Trade names and abbreviations

Benzophenone-2 (BP-2):

NSC-38556

Ref.: <https://drugs.ncats.io/drug/PRR8K3H9VN#names> consulted on 18 January 2025

Benzophenone-5 (BP-5):

Sulisobenzone sodium, UVISTAT 1121, UVIN 1MS40, ASL 24ST, sulibenzone

Ref.: <https://drugs.ncats.io/drug/853Z42ZYAS#names> consulted on 18 January 2025

3.1.1.4 CAS / EC number

Benzophenone-2 (BP-2):

CAS No. 131-55-5

EC No. 205-028-9

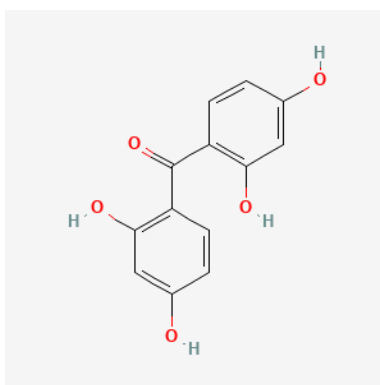
Benzophenone-5 (BP-5):

CAS No. 6628-37-1

EC No. 613-918-7

3.1.1.5 Structural formula

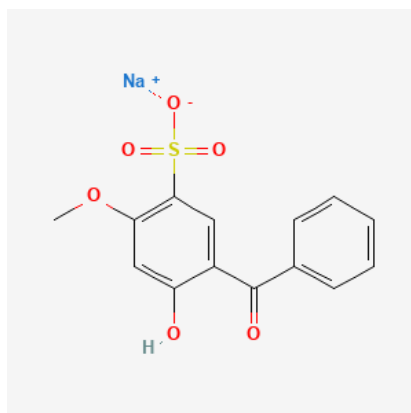
Benzophenone-2 (BP-2):



(from <https://pubchem.ncbi.nlm.nih.gov/compound/8571#section=Structures>, consulted on 27 September 2024)

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

Benzophenone-5 (BP-5):



(from <https://pubchem.ncbi.nlm.nih.gov/compound/23674885#section=2D-Structure>, consulted on 27 September 2024)

3.1.1.6 Empirical formula

Benzophenone-2 (BP-2):

$C_{13}H_{10}O_5$

(from <https://substances.ineris.fr/substance/131-55-5>)

Benzophenone-5 (BP-5):

$C_{14}H_{11}NaO_6S$

(from <https://echa.europa.eu/fr/substance-information/-/substanceinfo/100.121.934>)

3.1.2 Physical form

Benzophenone-2 (BP-2):

- Yellow needle-like crystals

(from Haynes, W.M. (ed.). CRC Handbook of Chemistry and Physics. 95th Edition. CRC Press LLC, Boca Raton: FL 2014-2015, p. 3-504, cited by <https://pubchem.ncbi.nlm.nih.gov/compound/8571#section=Computed-Properties>)

- Light yellow powder

(from Durairaj RB; Resorcinol: Chemistry, Technology and Applications. Springer Science & Business Media, p.557 (2005), cited by <https://pubchem.ncbi.nlm.nih.gov/compound/8571#section=Computed-Properties>)

Benzophenone-5 (BP-5):

- White or light-yellow powder

(from <https://cosmetics.specialchem.com/product/i-hangzhou-lingeba-technology-benzophenone-5>)

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

3.1.3 Molecular weight

Benzophenone-2 (BP-2):

246.21 g/mol, computed by PubChem 2.2 (PubChem release 2021.10.14)

Benzophenone-5 (BP-5):

330.29 g/mol, computed by PubChem 2.2 (PubChem release 2021.10.14)

(from <https://pubchem.ncbi.nlm.nih.gov/compound/23674885#section=Chemical-and-Physical-Properties>, consulted on 27 September 2023)

3.1.4 Purity, composition and substance codes

3.1.5 Impurities / accompanying contaminants

Benzophenone 2:

Pb: 8 ppm

As: 1 ppm

Maximum moisture content reported: 5%

Ref. CIR 1 report

SCCS comment

Only limited data on the heavy metal impurities of Benzophenone-2 are available. Information on the purity and impurities in the representative batches of both Benzophenone-2 and Benzophenone-5 on the market are not available.

3.1.6 Solubility

Benzophenone-2 (BP-2):

In water:

- 8,821 mg/L at 24.99 °C

Yalkowsky, S.H., He, Yan, Jain, P. Handbook of Aqueous Solubility Data Second Edition.
CRC Press, Boca Raton, FL 2010, p. 941

- 0.00195 mol/L = 480.1 mg/L (predicted average)

EPA Comptox:

<https://comptox.epa.gov/dashboard/chemical/properties/DTXSID5041306>

Also soluble in organic solvents such as methanol, ethanol, ethyl acetate, methyl ethyl ketone. Very soluble in [acetone](#), [diethyl ether](#), [ethanol](#)

From Haynes, W.M. (ed.). CRC Handbook of Chemistry and Physics. 95th Edition.
CRC Press LLC, Boca Raton: FL 2014-2015, p. 3-504

Solubility in [Ethyl acetate](#) 10, [methanol](#) 50, [methyl ethyl ketone](#) 22, [toluene](#) < 1 (all g/100 mL at 25 °C)

From Durairaj RB; Resorcinol: Chemistry, Technology and Applications.
Springer Science & Business Media, p.557 (2005)

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

From the references cited in
<https://pubchem.ncbi.nlm.nih.gov/compound/8571#section=Solubility>,
consulted on 27 September 2024

Benzophenone-5 (BP-5):

Solubility in water:

0.786 mol/L = 259.6 g/L (predicted average)

EPA Comptox: <https://comptox.epa.gov/dashboard/chemical/properties/DTXSID40216481>

504.9 g/L estimated by EPI WSKOW model within the EPISUITE platform

SCCS comment

The available information shows that Benzophenone-2 is slightly soluble in water, whereas Benzophenone-5 is freely soluble in water according to the solubility classification indicated in SCCS/1647/22.

(SCCS/1647/22 - Corrigendum 2, Chapter 3-2.7 Solubility, Table 2)

3.1.7 Partition coefficient (Log Pow)

Benzophenone-2 (BP-2):

2.64 (predicted average)

Ref.: EPA Comptox:

<https://comptox.epa.gov/dashboard/chemical/properties/DTXSID5041306>

Benzophenone-5 (BP-5):

-1.26 (predicted average)

Ref.: EPA Comptox:

<https://comptox.epa.gov/dashboard/chemical/properties/DTXSID40216481>

3.1.8 Additional physical and chemical specifications

Benzophenone-2 (BP-2):

Benzophenone-2 is prepared either by the reaction of hydroxybenzenes with benzyl hydroxide in the presence of a metal halide catalyst, or by the condensation of resorcinol with 2,4 dihydroxybenzoic acid in the presence of POCl₃ and ZnCl₂.

Ref: CIR report 1983

Organoleptic properties (colour, odour, taste if relevant): /

Melting point: 198.8 °C (from Haynes, W.M. (ed.). CRC Handbook of Chemistry and Physics. 95th Edition. CRC Press LLC, Boca Raton: FL 2014-2015, p. 3-504).

195°C Ref. CIR 2021

198-200 °C Sigma-Aldrich; Safety Data Sheet

201 °C (experimental): EPA-Comptox

Boiling point: 491 °C Zenker A et al.; J Chromat 1202: 64-74 (2008)

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

	442 °C (predicted average): EPA-Comptox
Flash point:	271 °C (predicted average): EPA-Comptox
Vapour pressure:	5.00e-9 (predicted average): EPA-Comptox
Density:	1.21 g/ cm ³ at 25 °C (from Durairaj RB; Resorcinol: Chemistry, Technology and Applications. Springer Science & Business Media, p.557 (2005))
	1.54 (predicted average): EPA-Comptox
Viscosity:	26.3 (predicted EPA-Comptox)
pKa	pKa1 =7.95, pKa2=6.99
pH:	/
Refractive index:	1.72 (predicted) Ref.: EPA-Comptox
UV/visible light absorption $\lambda_{\max}$:	283 nm.

SCCS comment

For Benzophenone-2, some values for physicochemical properties are predicted averages taken from [CompTox Chemicals Dashboard v2.5.1](https://comptox.epa.gov/dashboard/chemical/properties/DTXSID5041306) (EPA-Comptox)

Ref: U.S. Environmental Protection Agency. Comptox Chemicals Dashboard.
<https://comptox.epa.gov/dashboard/chemical/properties/DTXSID5041306> (accessed January 19, 2025) Sulibenzone.

Benzophenone-5 (BP-5):

Organoleptic properties (colour, odour, taste if relevant): /

Melting point:	227 °C (predicted average): EPA Comptox
Boiling point:	469 °C (predicted average): EPA Comptox
Flash point:	/
Vapour pressure:	3.98e-10 (predicted): EPA Comptox
Density:	1.52 g/cm ³ : EPA Comptox
Viscosity:	/
pKa1=	-2.4 pKa2= -5 (estimated with Chemaxon)
	https://go.drugbank.com/salts/DBSALT002279 Consulted on 18 January 2025
pKa:	pKa1 = 5.10, pKa2=-0.702 (predicted): EPA Comptox
PH:	6.0-6.5 www.lingebea.com/en/product/product-31-958.html
Refractive index:	/
UV/visible light absorption $\lambda_{\max}$:	286nm and 323nm.
Ref.:	Shaath N.A. Photochem. Photobiol. Sci., 2010, 9, 464–469
Ref.:	U.S. Environmental Protection Agency. Comptox Chemicals Dashboard
	https://comptox.epa.gov/dashboard/chemical/details/DTXSID40216481 (accessed January 19, 2025) Sulibenzone

SCCS comment

The majority of values for physicochemical aspects are the predicted values taken from US-EPA's CompTox Chemicals Dashboard for BP-5.

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5) (CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

3.1.9 Homogeneity and Stability

Benzophenone-2

Stable under standard ambient conditions (room temperature)

Ref.: Sigma-Aldrich; Safety Data Sheet for 2,2',4,4'-Tetrahydroxybenzophenone. Product Number: T16403, Version 6.8 (Revision Date 12/01/2024). Available from, as of January 18, 2025: <https://www.sigmaaldrich.com/GR/en/life-science/safety>

BP-2 should be stored in a closed system and be kept in a dark place without any light exposure.

Ref.: <https://www.lingebe.com/en/product/product-84-747.html>

Benzophenone-5

Should be stored in a closed system and be kept in a dark place without any light exposure.

Ref.: <https://www.lingebe.com/en/product/product-31-958.html>

SCCS comment

Data on the stability of both benzophenone-2 and benzophenone-5 under use conditions in cosmetic product are not available.

3.2 READ-ACROSS ASSESSMENT

In the absence of available data on BP-5, the SCCS has used read-across approach based on the data from other structurally similar analogues as explained below:

As the first step, a database within the OECD QSAR ToolBox (version 4.5 SP1) that contains >318,000 substances was searched to identify structural analogues of BP-5. For a reliable read-across, only the most closely similar analogues ($\geq 80\%$ similarity to the target compound) are used, because at lesser levels of similarity the molecular structures become more and more 'dissimilar' to the target compound. However, the SCCS carried out the initial search for up to 70% similarity to identify a wider range of analogues in the first step.

The search identified a total of 31 analogues. Two of these were duplicate entries and were therefore excluded, leaving a total of 29 analogues that had $\geq 70\%$ similarity to BP-5. These are listed in the Annex in a decreasing order of similarity.

In the second step, 4 analogues were selected that had $\geq 80\%$ similarity with BP-5 for use in read-across. These include Benzophenone-4 (CAS 4065-45-6); Sulisobenzone hydrate (the hydrated form of BP-4 with no separate CAS); Methanone, (2-hydroxy-4-methoxy-5-(sulfooxy)phenyl)phenyl- (CAS 84055-67-4); and 5-Bromo-2-hydroxy-4-methoxybenzophenone (CAS 3286-93-9) (Table 1).

In the third step, the OECD QSAR ToolBox database was searched for any existing experimental data to use in drawing estimates on the target substance (BP-5). The data matrix for the selected analogues (Table 1) showed that no experimental data were available for the 4 analogues – except for BP-4 for which data were found on acute toxicity; various

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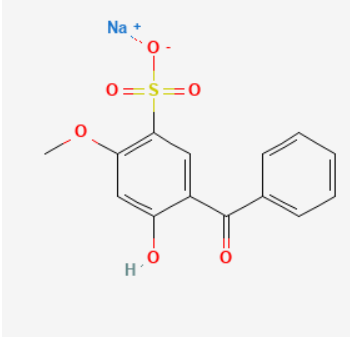
ToxCast assays for human health hazards; and *in vitro* relative binding to ER and AR receptors.

An analogue (not included in read-across) for which some experimental data were found is BP-3 (oxybenzone CAS 131-57-7). However, it is not structurally close enough to BP-5 (similarity 71.8%) to be considered for read-across.

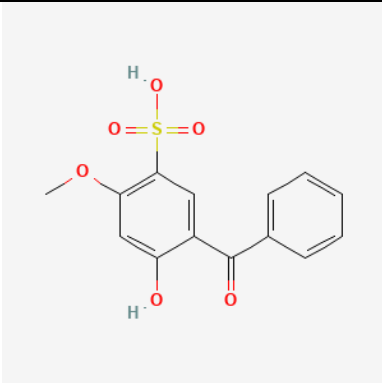
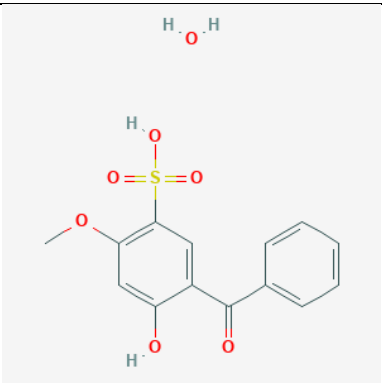
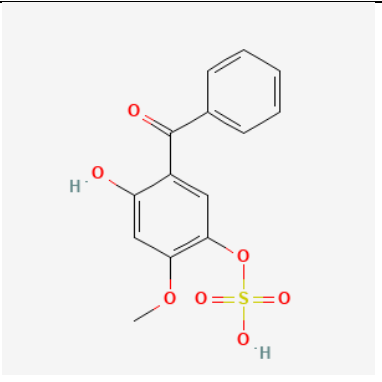
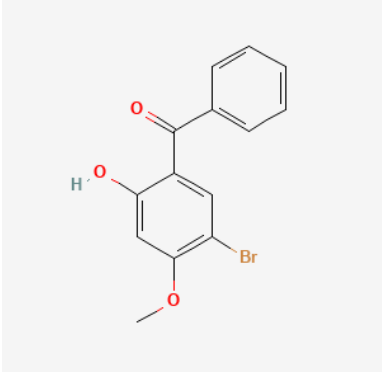
In view of this analysis, and the following considerations, the SCCS regarded it appropriate to use the available toxicological data on BP-4 for read-across to BP-5:

- BP-4 is the closest analogue to BP5. Both BP-5 and BP-4 have identical chemical structures – except that the sulfonic acid moiety in BP-4 exists in a sodium salt form in BP-5. Under polar (aqueous) solvent conditions, the ionisation of either sulphonic acid (in BP-4) or sodium salt of sulfonic acid (in BP-5) will generate the same anionic sulfonate moiety in both BP-4 and BP-5. It is also evident from the identification of structural analogues of BP-5 (Table 1) that the other 3 analogues are either a hydrated form of BP-4 (no specific CAS), or sulphonyl (CAS 84055-67-4) or bromo (CAS 3286-93-9) substituted variants of the core structure of BP-4. Thus, the overall comparison suggests that BP-4 is the closest analogue of BP-5 for use in read-across.
- Both BP-4 and BP-5 have similar physicochemical profiles. While BP-4 is freely soluble in water due to the presence of sulphonic acid moiety, the sodium salt form of sulphonic acid makes BP-5 even more water soluble. The experimental values for water solubility of BP-4 range between 300 and 314 g/L at 25°C (SCCS /1660/23), whereas a value of 504.9 g/L is estimated for BP-5 by the EPI WSKOW model within the EPISUITE platform.
- Both BP-4 and BP-5 have similar (QSAR-predicted) toxicological profiles (not included in this Advice).
- Because of the sulphonic acid moiety, both BP-4 and BP-5 are freely soluble in water and any systemically absorbed amounts can be expected to undergo a relatively fast clearance compared to other (hydrophobic) benzophenones. Although no experimental data on the water solubility of BP-5 are available, being the sodium salt form of BP-4, it can be expected to be even more water soluble, and hence of relatively less toxicological concern than BP-4.

Table 1: Analogues of BP-5 with ≥80% structural similarity to BP-5 identified from the OECD Toolbox for use in read-across

CAS number	SMILES	Name	Similarity (%)	Chemical Structure
6628-37-1	[Na+].COc1cc(O)c(cc1S([O-])(=O)=O)C(=O)c1ccccc1	Benzophenone-5 (The substance) target	100	

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4065-45-6	<chem>COc1cc(O)c(cc1S(=O)(=O)C(=O)c1ccccc1</chem>	Benzophenone-4 (Sulisobenzene)	88.3	
No specific CAS (same CAS used as of BP-4)	<chem>O.COc1cc(O)c(cc1S(=O)(=O)C(=O)c1ccccc1</chem>	Sulisobenzene hydrate (hydrated form of BP-4)	86.36	
84055-67-4	<chem>COc1cc(O)c(cc1OS(=O)(=O)C(=O)c1ccccc1</chem>	Methanone, (2-hydroxy-4-methoxy-5-(sulfooxy)phenyl)phenyl-	81.82	
3286-93-9	<chem>COc1cc(O)c(cc1Br)C(=O)c1ccccc1</chem>	5-Bromo-2-hydroxy-4-methoxybenzophenone	80.0	

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3.3 TOXICOKINETICS

3.3.1 Dermal / percutaneous absorption

Benzophenone 2

No data on dermal absorption are available. A default value of 50% as recommended in the SCCS Note of Guidance (SCCS/1647/22) will be used in the calculation of systemic exposure dose (SED).

Benzophenone 5

As explained in section 3.2, in the absence of data on dermal absorption of BP-5, the SCCS has considered it appropriate to use the same dermal absorption values as for BP-4 - the closest analogue of BP5 - for safety assessment of BP-5. In regard to dermal absorption, the values estimated by EPI DERMWIN model (v2.02) within EPISUITE platform also indicate a comparable profile for both BP-4 and BP-5 (5.22 µg/cm² and 7.32 µg/cm², respectively). Therefore, the SCCS used the same dermal absorption values for BP-5 as for BP-4, for which an OECD Test Guideline 428 *in vitro* skin penetration study from 2021 on human skin is available (SCCS /1660/23). The study reported that topical application of 2 mg/cm² radiolabelled BP-4 at 5% in a standard sunscreen application formula base resulted in a dermal delivery of 0.20 + 0.13% (Mean + 1SD) after 24h exposure. From this skin absorption study, the SCCS had used 0.46% (Mean + 2SD) for the calculation of SED for BP-4. This value will also be applied to BP-5 in this assessment.

3.3.2 Other studies on toxicokinetics

ADME

Human *In vitro*

Benzophenone-2

The fate of Benzophenone-2 was studied in human and zebrafish *in vitro* cell models (Le Fol *et al.*, 2015). In the human *in vitro* cell models, Benzophenone-2 was metabolised to a variety of gluco- and sulfo-conjugated metabolites. Similar patterns of Benzophenone-2 biotransformation were observed among zebrafish models (primary hepatocytes, ZFL, and ZELH-zfER cell lines). Metabolic patterns in the zebrafish models and human hepatic cell line HepaRG shared many similarities, while biotransformation rates in the cell lines MELN (human female cancer (invasive ductal carcinoma) cell line) and T47D-KBLuc (human female cancer (mammary gland breast/duct) cell line) used for estrogenicity testing were quantitatively low and qualitatively different. The biotransformation capability (bioactivation/detoxification processes) of Benzophenone-2 can influence its estrogenic potency (see section 3.5.10 below).

Animal Dermal

Benzophenone-2

A study was performed, using groups of 10 male Wistar rats, to determine the concentrations of Benzophenone-2 in the rat brain after topical administration (Broniowska *et al.*, 2019). Benzophenone-2 was dissolved in a small amount (volume not stated) of ethanol and olive oil and formulated with Hascobase (Hasco-Lek, Poland)⁶. The test substance was then applied to shaved skin at a dose of 100 mg/kg twice per day for 4 wks (days per wk not stated). Hascobase, with a small amount of ethanol and olive oil, was applied to the skin of control rats. Blood and tissue Benzophenone-2 concentrations in the frontal cortex and hippocampus were determined. After dermal application (24 h after last dose at 4 wks), the blood level of Benzophenone-2 was about 300 ng/ml. Liver and adipose tissue concentrations were 1354 ng/g wet tissue and 823 ng/g wet tissue, respectively. In the brain structures studied, the Benzophenone-2 concentration ranged from 5 to 19 ng/g tissue. In the hippocampus, the Benzophenone-2 concentration was approximately 3.5-fold lower in the frontal cortex. No significant changes in either oxidative stress or apoptosis markers were observed. There was also no change in apoptosis markers in the frontal cortex, but unexpectedly the oxidative stress markers were reduced.

To assess the concentration of total Benzophenone-2 (parent compound and its metabolites – glucuronide and sulfate), the liver was homogenized and plasma was mixed with 1 M ammonium acetate buffer. Prior to incubation of the homogenate for 6 h, freshly prepared enzyme mixtures (β -glucuronidase and sulfatase) were added. After hydrolysis with β -glucuronidase and sulfatase, the Benzophenone-2 peak was significantly higher than in the same serum and liver samples before hydrolysis. Calculation of the Benzophenone-2 concentration from the calculated standard curves revealed that the test compound was present in the plasma of treated animals at concentrations ranging from 164 to 648 ng/ml (average = 324 ng/ml; 1.3 μ M). After hydrolysis, the Benzophenone-2 concentration was 2218 ng/ml (9 μ M). These results indicated that, in the blood, there was more of the Benzophenone-2 metabolites than the parent compound. Additionally, in the liver, the Benzophenone-2 concentration after hydrolysis was much higher (3758 ng/g) when compared to the free form of the compound (1482 ng/g). Benzophenone-2 concentrations in all examined tissues in control animals were below the detection limit. The authors noted that the results of this study indicate that Benzophenone-2 passes through the blood-brain barrier, but that its concentration in the brain structures is much lower than in the blood.

The study showed that Benzophenone-2 did not exacerbate oxidative stress and apoptosis markers in the hippocampus and frontal cortex, and even lowered oxidative stress in the frontal cortex.

Animal Oral

Benzophenone-2

A dose-response experiment involving 5 doses (10, 33, 100, 333, or 1000 mg/kg) of Benzophenone-2 was performed using female Sprague-Dawley adult, ovariectomized rats (groups of 5) (Schlecht, 2008). Doses were administered (by gavage) once per day for 5 d. Blood and urine samples were collected at different time points (every 30 min) after test

⁶ The composition of Hascobase is described in another publication (Broniowska *et al.*, 2023) as paraffin liquid, white petrolatum, glycerol monostearate SE, cetostearyl alcohol, polysorbate 40, triglycerides of saturated fatty acids with an average chain length, propylene glycol, colloidal anhydrous silica, sorbic acid, water.

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substance administration. Additionally, the time-dependent metabolism and excretion of Benzophenone-2 were analyzed in a kinetic experiment, for further identification of metabolites. In this kinetic experiment, urine and serum samples were analyzed after *Helix pomatia* glucuronidase-/sulfatase (HPG) hydrolysis. Serum concentrations of Benzophenone-2 after dosing ranged from 0.1 µg/ml (after 10 mg/kg dose) to 1.1 µg/ml (after 1000 mg/kg dose). After hydrolysis with HPG, the serum concentrations of total Benzophenone-2 ranged from 1 to 62 µg/ml. Benzophenone-2 was metabolized to glucuronide- and sulfate-conjugates. In the serum, the maximum concentrations of Benzophenone-2, benzophenone-2-glucuronide, and benzophenone-2-sulfate were observed at 30 min post-dosing. The highest concentrations of Benzophenone-2 and its metabolites in the urine were measured at 120 min post-dosing. It was suggested that this biotransformation occurs via a first-pass effect in the gut wall or the liver.

SCCS comment

From this study, the SCCS concludes that a large proportion of the applied BP2 is rapidly metabolised and excreted.

Although data/information relating to ADME of BP-5 are not available, considering the high solubility in water, any systemically absorbed BP-5 can also be expected to be rapidly cleared and excreted via urine.

3.4 EXPOSURE ASSESSMENT

3.4.1 Function and uses

In the original CIR report (1983), Benzophenone-2 was the benzophenone with the highest reported use frequency (229 uses total); in 2021, use frequency of Benzophenone-2 decreased to 55 uses. It was reported that According to the industry's voluntary submission to the Food and Drug Administration (FDA) in 1976, Benzophenones are used in over a thousand cosmetic formulations, typically in concentrations up to 1%. The following were the maximum reported product concentrations for each Benzophenone: Benzophenone-1, 1%; Benzophenone-2, 5%; Benzophenone-3, 1%; Benzophenone-4, 10%; Benzophenone-5, < 0.1%; Benzophenone-6, 1%; Benzophenone-8, 1%; Benzophenone-9, 1%; Benzophenone-11, 5%. Benzophenones-7, -10, and -12 had no cosmetic use. Product formulation data for Benzophenones are listed in the Table below.

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Table 2: Benzophenone 2 - Product formulation data (from CIR report 1983)

Product category ^b	Total no. containing ingredient	No. product formulations within each concentration range (%) ^b				
		Unreported concentration	>5-10	>1-5	>0.1-1	≤0.1
Benzophenone-2						
Bath oils, tablets, and salts	3	—	—	—	—	3
Bubble baths	5	—	—	—	1	4
Other bath preparations	6	—	—	—	—	6
Colognes and toilet waters	120	—	—	1	27	92
Perfumes	22	—	—	—	1	21
Sachets	4	—	—	—	—	4
Other fragrance preparations	15	—	—	—	5	10
Hair conditioners	2	—	—	—	—	2
Hair rinses (noncoloring)	4	—	—	—	—	4
Hair shampoos (noncoloring)	14	—	—	—	2	12
Tonics, dressings, and other hair grooming aids	2	—	—	—	—	2
Wave sets	3	—	—	—	3	—
Blushers (all types)	3	—	—	—	—	3
Rouges	1	—	—	—	—	1
Makeup fixatives	1	—	—	—	—	1
Other makeup preparations (not eye)	4	—	—	—	—	4
Feminine hygiene deodorants	1	—	—	—	—	1
Aftershave lotions	30	—	—	—	6	24
Preshave lotions (all types)	1	—	—	—	—	1
Skin cleansing preparations (cold creams, lotions, liquids, and pads)	6	—	—	—	2	4
Face, body, and hand skin care preparations (excluding shaving preparations)	7	—	—	—	—	7
Moisturizing skin care preparations	8	—	—	—	—	8
Paste masks (mud packs)	1	—	—	—	—	1
Skin lighteners	1	—	—	—	—	1
Skin fresheners	27	—	—	—	—	27
Wrinkle smoothers (removers)	1	—	—	—	—	1
Skin care preparations	5	—	—	—	1	4
Suntan gels, creams, and liquids	1	—	—	—	—	1
Other suntan preparations	1	—	—	—	—	1
1976 TOTALS	299	—	—	1	48	250
1979 TOTALS^c	321	80	—	2	32	207

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SCCS comment

Only old (1983) data are available on the use of Benzophenone-2 and the concentrations used in different cosmetic products. In the absence of more recent data, the SCCS will use the available information in Table 2 and consider the use of Benzophenone-2 in all cosmetic products at the 1% maximum concentration, as only very few products had higher concentrations.

Table 3: Benzophenone 5 - Product formulation data (from CIR report 1983)

Product category ^b	Total no. containing ingredient	No. product formulations within each concentration range (%) ^b				
		Unreported concentration	>5-10	>1-5	>0.1-1	≤0.1
Benzophenone-5 Face, body, and hand skin care preparations (excluding shaving preparations)	7	–	–	–	–	7
Night skin care preparations	3	–	–	–	–	3
1976 TOTALS	10	–	–	–	–	10
1979 TOTALS ^c	11	–	–	–	–	11

Benzophenone-3, Benzophenone-4, and Benzophenone-5, but not the other benzophenones in the CIR safety assessment (2021), are included on the European Union's list of ultraviolet light (UV) filters allowed in cosmetic products.

European Commission. CosIng database; following Cosmetic Regulation No. 1223/2009. Last Updated 2020.

https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:02009R1223_20200501&from=EN Accessed 9-8-2020.

SCCS comment

Data on the use of Benzophenone-5, and the concentration used, in different cosmetic products are not available. However, since Benzophenone-5 is allowed in ready to use cosmetic preparations at concentrations up to 5% (as acid) (Annex VI/22 of the cosmetic regulation), the SCCS will use this concentration for the calculation of Systemic Exposure Dose (SED) for BP-5.

3.4.2 Calculation of SED/LED

Benzophenone-2 is not included on the European Union's list of ultraviolet light (UV) filters allowed in cosmetic products. Based on the Table 2 above, a concentration of 1% is used in this Advice to estimate the consumer exposure to BP2 via cosmetic products.

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Benzophenone-5 is allowed in ready for use cosmetic preparations at concentrations up to 5% (as acid). This concentration is therefore used in this Advice to estimate the consumer exposure to BP5 via cosmetic products.

For both benzophenones, the values of SED have been estimated following the recommendations in the SCCS Note of Guidance (SCCS/1647/22):

Summary of exposure to Benzophenone-2 at 1%:

Table 4: Summary of exposure to benzophenone-2 based when used in cosmetic products at 1%. The exposure scenarios have been considered the same as for BZ-4 in the SCCS opinion (SCCS/1660/23).

Product type	SED (mg/kg bw/day)
Dermal exposure:	
Sunscreen lotion*	1.5
Face cream*	0.12
Hand cream*	0.16
Dermal aggregate for leave-on and rinse-off products	1.5
Oral exposure:	
Lip stick*	0.009
Inhalatory exposure:	
Sunscreen propellant spray	0.018
Sunscreen pump spray	0.0002
Overall aggregate* (deterministic)	1.79

*Aggregate exposure has been calculated from face cream, hand cream, lipstick and sunscreen lotion (only one sunscreen was considered for the worst-case scenario)

Summary of exposure to Benzophenone-5 at 5%:

Table 5: Summary of exposure to benzophenone-5 based on read-across from BP-4 when used in cosmetic products at 5%. The exposure scenarios have been considered the same as for BZ-4 in the SCCS opinion (SCCS/1660/23).

Product type	SED (mg/kg bw/day)
Dermal exposure:	
Sunscreen lotion	0.069
Face cream*	0.006
Hand cream*	0.008
Dermal aggregate for leave-on and rinse-off products	0.067
Oral exposure:	
Lip stick*	0.045
Inhalatory exposure:	
Sunscreen propellant spray*	0.088
Sunscreen pump spray	0.001
Overall aggregate* (deterministic)	0.147

*Aggregate exposure has been calculated from face cream, hand cream, lipstick and sunscreen propellant spray (only one sunscreen was considered for the worst-case scenario)

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3.5 TOXICOLOGICAL EVALUATION (BP-2)

To keep a continuity of the assessment, the toxicological data have been evaluated for BP-2 in this section. The toxicological evaluation for BP-5 is given separately in section 3.6.

3.5.1. Irritation and corrosivity

3.5.1.1 Skin irritation

Procedures outlined by the Federal Hazardous Substances Labeling Act (FHSLA) were used to test Benzophenones for acute skin irritation. An occlusive patch containing 0.5 ml or 0.5 mg of the test ingredient was applied to the intact and abraded skin of albino rabbits. Patches remained in place for 24 hours and were then removed and scored for irritation according to the Draize method. Sites were again scored 24 hours after patch removal. Benzophenones-2, was non-irritating to intact and abraded skin when tested at concentrations from 4% to 100%.

Ref: IBL. (Aug. 1964).

Submission of data by CTFA. Unpublished data, both cited in the CIR report 1983

3.5.1.2 Mucous membrane irritation / eye irritation

A number of studies have determined the potential irritancy of Benzophenones to the eyes of rabbits. The test material (0.1 ml or 0.1 g) was instilled into one eye of each rabbit; the other eye served as an untreated control. Eyes were examined and scored for irritation daily for a period of three to ten days. Some test procedures included washing of the treated eyes with water four seconds after instillation of the test material. Results of eye irritation tests revealed that most Benzophenones at concentrations of 5%-100% were non-irritating when instilled into the eyes of rabbits.

Benzophenone-2, was slightly to moderately irritating at 100% concentration; however, it was non-irritating when tested at 16% in dimethyl phthalate (DMP) or petrolatum.

Ref: IBL. (Oct. 1965). Submission of data by CTFA. Unpublished data, both cited in the CIR report 1983

3.5.2 Skin sensitisation

In vitro or animal data are not available for Benzophenone-2.

Human data:

There are sporadic reports on sensitisation to Benzophenone-2 (Boehnke *et al.*, 1997).

A female patient had skin reactions on her fingers, with a positive patch-test reaction to a nail polish remover, suggested to contain Benzophenone-2.

A male patient presented with subacute chest and arm eczema after use of a toilet water product. Patch testing with an ingredient of the product, Benzophenone-2, showed a positive reaction (Jacobs, 1998).

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Severe dermatitis (worsened after sun exposure) was observed in a female patient. Patch test results for Benzophenone-2 were positive (Gimenez-Arnau *et al.*, 2002).

SCCS comment

In view of only a limited number of reported cases in the literature from the potential exposure of a large number of individuals, the SCCS considers that risk of sensitisation to Benzophenone-2 can be regarded as low.

3.5.3 Acute toxicity

3.5.3.1 Acute oral toxicity

Benzophenone-2, is practically nontoxic to slightly toxic when administered orally to rats at doses up to 16 g/kg (LD50 = 1.22 g/kg; convulsion and immediate death at highest dosage ie 3.5g/kg in corn oil at 5%).

Ref: INDUSTRIAL BIOLOGY LABS (IBL). (July 1964). Submission of data by CTFA.
Unpublished data cited in the CIR 1 report (1983, Ref 100)

3.5.3.2 Acute dermal toxicity

/

3.5.3.3 Acute inhalation toxicity

/

3.5.4 Repeated dose toxicity

/

3.5.4.1 Repeated dose (28 days) oral / dermal / inhalation toxicity

/

3.5.4.2 Sub-chronic (90 days) oral / dermal / inhalation toxicity

/

3.5.4.3 Chronic (> 12 months) toxicity

/

3.5.5 Reproductive toxicity

3.5.5.1 Fertility and reproduction toxicity

/

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3.5.5.2 Developmental Toxicity

The developmental toxicity of Benzophenone-2 (in 10% ethanol/90% corn oil vehicle) was evaluated using groups of 5 timed pregnant C57BL/6NCr mice (Hsieh et al., 2007). Benzophenone-2 (6.25 mg) was administered via gavage from gestational days 12 through 17. They were sacrificed on day 18. Fetuses were weighed and sexed, anogenital distance was measured and genital tubercles were harvested for paraffin sections or quantitative reverse transcriptase-polymerase chain reaction analysis of genes purportedly involved in genital tubercle development.

In the test group, 8 of 57 male fetuses had hypospadias ($p = 0.0064$, when compared to controls). The co-administration of Benzophenone-2 with an estrogen receptor antagonist (10 μg in vehicle (subcutaneously (s.c.)) during gestation, yielded normal genital tubercles; *i.e.*, no hypospadias in 26 of 26 mice. Likewise, no EM-800 or control treated male genital tubercles showed hypospadias. Benzophenone-2 treated male mice had no changes in body mass adjusted anogenital distance relative to controls. Reverse transcriptase-polymerase chain reaction revealed that genital tubercles of benzophenone-2 treated male mice expressed higher levels of estrogen receptor- β relative to male controls ($p = 0.04$).

The authors concluded that Benzophenone-2 may cause hypospadias via signalling through the estrogen receptor. Further study of human benzophenone-2 exposure and its effects is needed to support this hypothesis.

3.5.6 Mutagenicity / genotoxicity

3.5.6.1 Mutagenicity / genotoxicity *in vitro*

Benzophenone-2 (tested up to 10,000 $\mu\text{g}/\text{plate}$), was reported to be weakly mutagenic with metabolic activation in the Ames test. Benzophenone-2 was weakly mutagenic in the mouse lymphoma forward mutation assay (at doses of 24 and 32 $\mu\text{g}/\text{plate}$) and in a cytogenetic assay evaluating sister chromatid exchanges and chromosome aberrations (at doses of 100 and 200 $\mu\text{g}/\text{plate}$). These effects in L5178Y mouse lymphoma cells were only observed at the high end of the range of doses tested.

In a modified Ames test, Benzophenone-2 (concentrations up to 1000 $\mu\text{g}/\text{ml}$) was not genotoxic to *Salmonella typhimurium* strains with or without metabolic activation (Elder, 1983). Benzophenone-2 did not induce unscheduled DNA synthesis in rat hepatocytes at concentrations up to 1000 nmol/ml.

The mutagenicity of Benzophenone-2 was investigated with the Ames *Salmonella* /mammalian-microsomal assay. The strains TA98, TA 100, TA1535, TA1537, and TA1538 were used; all tests were performed in the presence and absence of Aroclor-induced liver microsomal S-9 cell fraction to observe the mutagenic effect of each compound following metabolic activation. Preliminary cytotoxicity studies determined the dose range of each compound to be used. Benzophenone-2 was non-mutagenic when assayed directly and was mutagenic with metabolic activation in *Salmonella* strain TA1537. Benzophenone-2, in the presence of rat liver microsomes induced a "small but fairly consistent increase in the number of mutants" in four *Salmonella* strains tested. At doses of 100-300 μg , Benzophenone-2 induced mutant increases of 50-100% in TA100 and 200-500% in TA 1537. A mutant increase of 50% was observed in strains TA98 and TA1535, but these strains have not been tested for long enough time to provide conclusive results. The investigator suggested that the "small and somewhat erratic nature of the mutagenic response may be due to the presence of an impurity." The purity of the test sample was 99% (Lab grade) and was assumed to be purer

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than that of the cosmetic-grade. Additional tests using lab-grade Benzophenone-2 found this ingredient to be mutagenic in TA1537 at doses of 200 and 750 µg when activated by Aroclor induced hamster liver enzymes. Preliminary assays of cosmetic-grade Benzophenone 2 revealed mutagenic activity not differing significantly from that of the purer lab-grade

Ref. 1-5 in CIR, 1983:

1: *Cosmetic Toiletry, and Fragrance Association (CTFA). 1980c. Submission of data by CTFA. Supplementary mutagenesis data. (not made available to SCCS).*

2: *Department of Health, Education and Welfare (DHEW). May 1978. Submission of data by CTFA. Unpublished data on Benzophenone, Appendix 5b.*

3: *DHEW. Feb 1979. Submission of data by CTFA. Unpublished data. Appendix 5c.*

4: *HILL TOP RESEARCH LABS, June 1979a. Submission of data by CTFA. Unpublished data. Appendix 3a.*

5: *HILL TOP RESEARCH LABS, Sept 1979b. Submission of data by CTFA. Unpublished data. Appendix 3b.*

Benzophenone-2 was reported to be positive for strains TA98, TA100, TA1535, and TA1537 with S-9 metabolic activation based on textual comments on the unpublished data submitted by industry to CIR for their report. The interpretation of these data, which have since been published, was clarified by the investigators who reported that Benzophenone-2 is "clearly mutagenic" only in strain TA1537 (ref. 3). It was suggested that this may indicate that Benzophenone-2 causes frameshift mutations by intercalating between DNA bases without covalent bonding, and that caution should be taken in extrapolating from mutagenicity to carcinogenicity test of such agents in the Ames test.

An *in vitro* cytogenic assay was used to evaluate the ability of Benzophenone-2 to induce sister chromatid exchange (SCE) and chromosome aberrations (CA) in L5178Y mouse lymphoma cells. Assays were performed in the presence and absence of Aroclor-induced rat liver microsomal enzymes (S9). The solubility of Benzophenone-2 in DMSO and its cytotoxicity were first determined. For the mutagenesis assays, doses of 6.250-200.00 µg Benzophenone-2 per plate were used. When assayed in the absence of S-9, Benzophenone-2 induced small "biologically insignificant" increases in SCE at 100 and 200 µg; CA frequencies were not elevated at any dose. With metabolic activation, however, Benzophenone-2 produced "statistically and biologically significant" increases in SCE frequency at the three highest dose levels indicating a dose-response relationship. The author noted that Benzophenone-2 was more toxic to cells with the activation system; only 17 scorable cells were found at the 100-µg dose. The investigators reported that ten CAs (including a quadriradial, a translocation, and two triradials) were observed among the 67 cells scored at the two highest doses with activation. He concluded that Benzophenone-2 does not directly induce significant SCE or CA increases but does, under metabolic activation, induce these changes. Ref 7 in CIR, 1983)

Ref 7: *LITTON BIONETICS. 1979. Submission of data by CTFA. Supplementary metagenesis data.*

A mouse lymphoma forward mutation assay was used to test Benzophenones-2 for mutagenesis. The L5178Y TK+/- cell in line was used; assays were performed in the presence and absence of Aroclor-induced rat liver microsomal preparation (S-9). Materials were dissolved in DMSO and tested for preliminary cytotoxicity to determine doses to be used in the assays. Without activation Benzophenone-2 was mutagenic at "highly toxic" doses. In the

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presence of S-9, Benzophenone-2 became more toxic. An increase in the mutant frequency (3.0-6.8 times) was observed with the three most toxic doses. A dose-response relationship was not demonstrated. The investigator suggested that "appears to react with microsomal system to yield a mutagenic product that induces mutants at lower applied concentrations and toxicities than under nonactivation conditions." It was concluded that Benzophenone-2 induced an increase in mutations at the TK locus in L5178Y mouse lymphoma cells only for highly toxic doses with or without metabolic activation and that this material is weakly mutagenic under the conditions of the test (Ref 7). These findings need to be reconfirmed since there was no dose response pattern of toxicities over the preferred relative growth range in any of the trials; increases in mutant frequency in the assays occurred only at level bordering total lethality; and the lethal dose was poorly reproduced from one trial to another with metabolic activation.

The gradient plate test of McMahon *et al.* (Ref 8 in the CIR report, 1983), a modification of the Ames test, was used to investigate the effects of Benzophenone-2 on bacterial mutation. (Ref 9, in the CIR report, 1983). The chemical was dissolved in DMSO and tested at concentrations 0.1-1000 µg/ml with and without rat liver S-9 metabolic activation. Benzophenone-2 was tested with strains C46, C 3076, TA98, TA 100, TA1535, TA1537, and TA1538 and Escherichia coli strains WP2 and WP2 uvrA-(tryptophan auxotrophs). Benzophenone 2 inhibited the growth of all the bacterial strains but did not induce mutations with or without metabolic activation.

Ref 8: McMAHON, R.E., CLINE, J.C., and THOMPSON, C.Z. of (1979). Assay of 855 test chemicals in ten tester strains using a new modification of the Ames test for bacterial mutagens. Cancer Res. 39, 682-93.

Ref 9: LILLY RESEARCH LABS. (May 1982a). Submission of data by CTFA. The effect of benzophenone-2 on the induction of bacterial mutation using a modification of the Ames test, Study 820510GPA1876.

The induction of unscheduled DNA synthesis (repair synthesis) in primary cultures of adult rat hepatocytes was studied after exposure of the cultures to Benzophenone 2 dissolved in DMSO and at concentrations of 0.5-1000 nmoles/ml (Ref 11 in CIR, 1983). Cytotoxicity was observed at the 500 and 100 nmoles/ml concentrations of Benzophenone 2. Benzophenone 2 did not induce DNA repair synthesis.

Ref 11: LILLY RESEARCH LABS. (June 1982c). Submission of data by CTFA. The effect of benzophenone-2 on the induction of DNA repair synthesis in primary cultures of adult rat hepatocytes, Study 82051 IUDS1876.

3.5.6.2 Mutagenicity / genotoxicity *in vivo*

Female Chinese hamsters were orally administered 62.5-500 mg/kg Benzophenone 2 in 10% Acacia solution (Ref 13 in CIR, 1983). The animals were sacrificed, and the frequency of bone marrow SCE exchange was determined. Benzophenone 2 did not induce SCE *in vivo* in the bone marrow of Chinese hamsters. Cytotoxicity was not observed.

Ref 13: LILLY RESEARCH LABS. (May 1982e). Submission

Overall comment of the SCCS on genotoxicity/mutagenicity of BP-2

Most of the data from genotoxicity/mutagenicity testing of BP-2 analysed by the SCCS came from study reports to which the SCCS had no access. The data were mostly cited in papers published in the 1980s. As the SCCS had no access to the original study reports, it relied on the assessment of results as evaluated by the authors of the studies. It is notable that one study on *in vitro* SCE reported positive results, while an *in vivo* SCE study reported negative results. Other genotoxicity endpoints such as *in vitro* Comet assays, or *in vitro* UDS indicate negative results.

After analysis of all the available data (summarised in Table 6 below) the SCCS has concluded that genotoxicity/mutagenicity potential of BP-2 cannot be excluded due to:

- positive results reported in gene mutations tests in different bacterial strains (Ames test) mostly with metabolic activation (*i.e.* +S9-mix),
- positive results reported in gene mutations in mammalian cells (Mouse Lymphoma Assay) with metabolic activation (*i.e.* +S9-mix),
- positive results reported in one chromosomal aberration test in mouse lymphoma L5178Y cells with metabolic activation (*i.e.* +S9-mix).

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Table 6: Summary of data on genotoxicity/mutagenicity testing of BP-2 available to the SCCS.

	Test/endpoint	Concentration tested	Result	Reference
	GENE MUTATIONS IN BACTERIA			
1	Ames test - modification of gradient plate test by McMahon: Salmonella strains: G46, C3076, D3052, TA98, TA100, TA1535, TA1537, TA1538, and Escherichia coli strains WP2 and WP2 uvrA-	BP-2 purity unknown 0.1-1000 µg/mL -/ +S9	According to author of the report: Negative -/+S9 Growth inhibition of all the bacterial strains but no mutations induced.	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LILLY RESEARCH LABS. (May 1982a). Submission of data by CTFA. The effect of benzophenone-2 on the induction of bacterial mutation using a modification of the Ames test, Study 820510GPA1876. (<i>not available to SCCS</i>)
2	Ames test - Salmonella strains: TA98, TA100, TA1535, TA1537	BP-2 purity unknown 0.1-10000 µg/mL -/ +S9 in DMSO	According to author of the report: Negative -S9 Positive +S9: clearly mutagenic in TA1537 +S9 Positive +S9 for all 4 strains according to JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 35-77. Mary Ann Liebert, Inc., Publishers	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following references were cited in the paper: 1. Department of Health, Education and Welfare (DHEW). May 1978. Submission of data by CTFA. Unpublished data on Benzophenone, Appendix 5b. (<i>not available to SCCS</i>) 2. DHEW. Feb 1979. Submission of data by CTFA. Unpublished data. Appendix 5c. (<i>not available to SCCS</i>) 3. Prival, M. J., Sheldon, A. T., & Popkin, D. (1982). Evaluation, using Salmonella typhimurium, of the mutagenicity of seven chemicals found in cosmetics. Food and Chemical Toxicology, 20(4), 427-432. doi:10.1016/s0278-6915(82)80108-7 The following papers/documents cite the data: 1: Amended Safety Assessment of Benzophenones as Used in Cosmetics. CIR. Final Amended Report Release Date: April 6, 2021.
3	Ames test - Salmonella strains: TA98, TA100, TA1535, TA1537, TA1538	BP-2 purity unknown 10-1000 µg/mL -/ +S9 in DMSO	According to author of the report: Negative -/+ S9	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. HILL TOP RESEARCH LABS, June 1979a. Submission of data by CTFA. Unpublished data. Appendix 3a. (<i>not available to SCCS</i>)
4	Ames test - Salmonella strains: TA98, TA 100, TA1535, TA1537	2,2',4,4'-Tetrahydroxybenzophenone (THBP, CAS No. 131-55-5) from Aldrich Chemical Co., purity 99%	Negative -S9 Positive +S9: "small but fairly consistent increase in the number of mutants" in four Salmonella strains tested.	Prival, M. J., Sheldon, A. T., & Popkin, D. (1982). Evaluation, using Salmonella typhimurium, of the mutagenicity of seven chemicals found in cosmetics. Food and Chemical Toxicology, 20(4), 427-432. doi:10.1016/s0278-6915(82)80108-7

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		3, 10, 30, 100, 300, 1000 µg/plate -/+S9 from rat liver	Positive: TA1537 +S9 (100-300 µg/plate): number of mutants increased by 200-500% Equivocal: TA100 +S9 (100-300 µg/plate): number of mutants increased by 50-100% Aroclor 1254-induced hamster liver S-9 was also capable of activating THBP to a mutagen on strain TA1537 (data not shown).	
5	Ames test - Salmonella strain: TA1537	BP-2 purity and concentration range unknown	According to author of the report: Positive +S9: at 200 and 750 µg/plate with S9 from hamster liver	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. HILL TOP RESEARCH LABS, Sept 1979b. Submission of data by CTFA. Unpublished data. Appendix 3b. (<i>not available to SCCS</i>)
GENE MUTATIONS IN MAMMALIAN CELLS				
6	Mouse Lymphoma Assay L5178Y cells	BP-2 purity unknown Concentration range not clear	According to author of the report: Positive -S9; at highly toxic doses Positive +S9: (3.0-6.8 times) observed with the three most toxic doses. A dose response relationship was not demonstrated. Increases in mutant frequency in the assays occurred only at level bordering total lethality; and the lethal dose was poorly reproduced from one trial to another with metabolic activation.	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LITTON BIONETICS. 1979. Submission of data by CTFA. Supplementary mutagenesis data. (<i>not available to SCCS</i>)
CHROMOSOMAL ABERRATIONS				
7	Chromosomal aberrations Mouse lymphoma L5178Y cells	BP-2 purity unknown 6.25-200 µg/plate -/+S9	According to author of the report: Negative -S9 Positive +S9	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LITTON BIONETICS. 1979. Submission of data by CTFA. Supplementary mutagenesis data. (<i>not available to SCCS</i>)
8	Micronucleus test BP-2 +/- UVA irradiation Human keratinocyte HaCaT cells	BP-2 from Sigma Chemical Co., purity unknown 25 µg/mL	Study not reliable, only poor quality images were provided to justify the statement by the Authors: "Micronuclei formation was detected in HaCaT cells treated with BP 2 (25 µg/mL) under UVB (1.08 J/cm ²)	Amar SK <i>et al.</i> Combined effect of Benzophenone-2 and ultraviolet radiation promote photogenotoxicity and photocytotoxicity in human keratinocytes. Regul Toxicol Pharmacol. 2018 Jun;95:298-306. doi: 10.1016/j.yrtph.2018.04.003.

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			exposure, simultaneously there was no micronuclei detected in control cells of UV and dark".	
	OTHER ENDPOINTS			
9	Unscheduled DNA Synthesis primary cultures of adult rat hepatocytes	BP-2 purity unknown 0.5-1000 nmoles/mL	According to author of the report: Negative Cytotoxicity at 500 and 100 nmoles/mL, but no induction of DNA repair synthesis	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LILLY RESEARCH LABS. (June 1982c). Submission of data by CTFA. The effect of benzophenone-2 on the induction of DNA repair synthesis in primary cultures of adult rat hepatocytes, Study 820511UDS1876. <i>(not available to SCCS)</i>
10	Sister Chromatid Exchanges <i>in vitro</i> Mouse lymphoma L5178Y cells	BP-2 purity unknown 6.25 - 200 µg/plate	According to author of the report: Negative -S9 Positive +S9 -S9: small "biologically insignificant" increases at 100 and 200 µg +S9: statistically and "biologically significant" increases in SCE frequency at the three highest dose levels indicating a dose-response relationship	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LITTON BIONETICS. 1979. Submission of data by CTFA. Supplementary mutagenesis data. <i>(not available to SCCS)</i>
11	Comet assay <i>in vitro</i> Human keratinocyte HaCaT cells	BP-2 purity unknown 5-25 µg/mL	In the dark conditions the BP-2 concentration-DNA damage relationship was observed, but no statistical significance reported. Study of limited reliability due to insufficient details on methodology (content of buffers, number of cells scored, or number of experiments).	Amar SK <i>et al.</i> . Combined effect of Benzophenone-2 and ultraviolet radiation promote photogenotoxicity and photocytotoxicity in human keratinocytes. Regul Toxicol Pharmacol. 2018 Jun;95:298-306. doi: 10.1016/j.yrtph.2018.04.003.
	IN VIVO TESTING			
12	Sister Chromatid Exchanges <i>in vivo</i> bone marrow cells of female Chinese hamsters	BP-2 purity unknown 62.5-500 mg/kg	According to author of the report: Negative Cytotoxicity was not observed. No increase in frequency of SCE.	JOURNAL OF THE AMERICAN COLLEGE OF TOXICOLOGY Volume 2, Number 5, 1983. Pages: 79-84. Mary Ann Liebert, Inc., Publishers The following reference was cited in the paper: 1. LILLY RESEARCH LABS. (May 1982e). Submission of data by CTFA. The effect of benzophenone-2 on the <i>in vivo</i> induction of sister chromatid exchange in bone marrow of Chinese hamsters, Study 820504SCE1876. <i>(not available to SCCS)</i>

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3.5.7 Carcinogenicity

Benzophenone-2:

Guideline compliant dermal or oral carcinogenicity studies are not available for BZ-2. As genotoxicity cannot be excluded, the SCCS considers that the concerns for carcinogenicity cannot be ruled out.

3.5.8 Photo-induced toxicity

3.5.8.1 Phototoxicity / photo-irritation and photosensitisation

Benzophenone-2:

Cosmetic products containing Benzophenones-2, -3, or -4 (0.1% to 3.5%) were evaluated for phototoxicity using human subjects. Products containing Benzophenones-2, -3, and -4 were non-phototoxic in all studies; however, several subjects experienced slight irritation (usually a 1 + response) to the test material (CIR, 2021).

Twenty-seven patients reported reactions due to sunscreen allergy (itchy bumps and burning). Of these, 11 (10 women, 1 man) patients agreed to photopatch testing. One patient had a delayed-type hypersensitivity photopatch test reaction to Benzophenone-2 (1% in petrolatum), and another patient had a photopatch test reaction to Benzophenone-3 (10% in petrolatum) (Shaw *et al.*, 2010).

SCCS comment

In view of the limited number of reported cases in the literature from the potential exposure of a large number of individuals, the SCCS considers the risk of photo-induced toxicity of Benzophenone-2 as low.

In this regard, the SCCS found a published study (Amar *et al.*, 2018) on phototoxicity of BP-2 in which human keratinocyte HaCaT cells were exposed to the single compound in dark condition (5–25 µg/mL) and after irradiation to UVA (2.7 J/cm²) or UVB (1.08 J/cm²). The cells were investigated in the MTT & NRU tests, as well as by EtBr and AO staining, mitochondrial transmembrane potential by JC-1 staining, and in flow cytometry to estimate apoptotic population by propidium iodide staining. The study results indicate increased cell damage by BP-2 after co-exposure to UVA (2.7 J/cm²) or UVB (1.08 J/cm²) irradiation.

The SCCS considers this study of limited reliability for consideration in this assessment because:

- the doses of UVA and UVB are not justified,
- the UVB dose is excessively high, and it does not correspond to typical sun irradiation intensity,
- BP-2 purity is unknown,
- no reference to whether any OECD or other Guideline on phototoxicity was followed.

3.5.8.2 Photomutagenicity / photoclastogenicity

The study by Amar *et al.* (2018) also tested photogenotoxicity of BP-2 in which human keratinocyte HaCaT cells were exposed to the single compound in dark condition (5–25 µg/mL) and after irradiation to UVA (2.7 J/cm²) or UVB (1.08 J/cm²). The cells were investigated in the Comet assay and micronucleus test. In the dark conditions the BP-2 concentration-DNA damage relationship was observed, but no statistical significance reported. The cells exposed to different concentrations of BP-2 under UVA (2.7 J/cm²) or UVB (1.08

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J/cm²) exhibited significant DNA damage compared to control cells. Micronuclei formation was detected in HaCaT cells treated with BP 2 (25 µg/mL) under UVB (1.08 J/cm²) exposure, simultaneously there was no micronuclei detected in control cells of UV and dark.

SCCS comment

The SCCS considered the study by Amar *et al.* (2018) as of limited reliability for use in safety assessment of BP-2, due to unknown purity of the test item and to the insufficient description of the micronucleus data as only poor-quality images are provided to justify the stated micronucleus induction. Conversely, the positive findings in the Comet assay are indicators of DNA damage that can contribute to the weight of evidence approach, but cannot be used as standalone tests (SCCS Notes of Guidance, 2023).

SCCS overall conclusion on Photomutagenicity/ photoclastogenicity

Overall, the available evidence is not sufficient for excluding potential photomutagenic effect of BP-2.

3.5.9 Human data

Biomonitoring Studies

Benzophenone-2

Benzophenone-1, Benzophenone-2, Benzophenone-3, and 4-hydroxybenzophenone (metabolite of Benzophenone-1 and Benzophenone-3) were detected in urine samples obtained from 20 male subjects (Vela-Soria *et al.*, 2014). The authors noted that there seemed to have been a relationship between the presence of Benzophenone-3 and Benzophenone-1 because, in all samples in which Benzophenone-3 was present, Benzophenone-1 was also present. Furthermore, they noted that this observation is suggestive of the possible conversion of Benzophenone-3 to Benzophenone-1 and that the content of Benzophenone-1 may be due to human metabolism to and not direct exposure.

Spot urine samples (157 total) obtained from a segment of the general German population (59 females, 39 males, and 59 children) were analyzed. Benzophenone-1 and Benzophenone-3 had high detection rates (26%).

A study was performed to investigate the exposure of human embryos and fetuses to UV filters. Placentas (12) from volunteer mothers in Spain were collected at delivery. Benzophenone-1 was detected in all samples. 4-Hydroxybenzophenone, metabolite of Benzophenone-1 and Benzophenone-3, was detected in 3 of the 12 placental samples. Urinary concentrations of benzophenones were measured in 34 Tunisian women. Benzophenone-1 and Benzophenone-3 were found in 91.2% and 64.7% of the analyzed samples, respectively.

Urinary concentrations of the following benzophenone derivatives were evaluated in a national sample of the South Korean population (1576 subjects) (Kang *et al.*, 2016). Benzophenone-1, Benzophenone-2, Benzophenone-3, Benzophenone-4, Benzophenone-8, and 4-hydroxybenzophenone. The detection rates for Benzophenone-1 and 4-hydroxybenzophenone were 56% and 88%, respectively. The detection rate for benzophenone-2 was below 14%: Benzophenone-2, Benzophenone-3, Benzophenone-4, and Benzophenone-8.

In the discussion, the authors mentioned that generally, the BP derivatives approved globally for use as UV filters, BP-1 and BP-2, are used as UV stabilizers in food packaging (Shaath, 2010).

BP-3, BP-4, and BP-8 are approved by the FDA for use as UV filters in sunscreens (Pathak and Fitzpatrick *et al.*, 1993). However, the concentrations of BP-2, BP-4, and BP-8 were

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virtually undetectable in this study – possibly reflecting their limited use compared to BP-1 and BP-3 – and were thus not considered. BP-2 and BP-8 were detected in only 5.6% and 2.6% of urinary samples in the USA (Kunisue *et al.*, 2012). The detection rates of BP-2 and BP-8 were 39.4% and 10.1% in urinary samples from young Chinese adults (Gao *et al.*, 2015). The study from Kang *et al.* (2016) showed that detection rate of BP-2, BP-4 and BP-8 (below 14%) was similar to the concentration reported in other countries. A possible explanation for the discrepancy in our study may be because, Korean population are generally exposed to BP-1 and BP-3. These concentrations in urine may be related to the variety factors, including exposure sources, route, and metabolic differences.

3.5.10 Special investigations

Endocrine Activation

In vitro

Estrogenic Effects

A direct interaction of BP2 with the estrogen receptor (ER) has been demonstrated *in vitro* with ligand binding assays (Seidlova-Wuttke *et al.*, 2004,) and reporter gene studies (Miller *et al.*, 2001; Yamasaki *et al.*, 2003).

Effects on Thyroid

In addition, BP2 might affect endocrine systems via interference with enzyme activity. At least *in vitro*, BP2 is a strong inhibitor of the thyroid peroxidase (TPO) reaction (Schmutzler *et al.*, 2007). However, whether BP2 disturbs thyroid hormone homeostasis *in vivo* by inhibiting or inactivating TPO remains to be determined.

A study was performed to investigate the thyroid-activation potential of benzophenones, using a rat pituitary carcinoma cell line (GH3 cell line) and a rat thyroid follicular cell line (FRTL-5 cell line) (Lee, 2018). The test concentrations in GH3 cells were as follows: Benzophenone-2 (up to 2.5 mg/l (10 µM)). In FRTL-5 cells, the test concentrations were Benzophenone-2 (up to 78.8 mg/l).

The results indicated that, in GH3 cells, Benzophenone-2 (0.32 to 10 µM) statistically significantly down-regulated the Tshβ, Trhr, and Trβ genes. Additionally, Benzophenones-2 (10 to 320 µM) statistically significantly upregulated the Nis and Tg genes, while down-regulating the Tpo gene in the FRTL-5 cells.

Whether or not Benzophenone-2 inhibits key reactions of thyroid hormone biosynthesis catalyzed by thyroid peroxidase was also examined in the study of Schmutzler *et al.*, (2007)(A novel *in vitro* assay, based on human recombinant thyroid peroxidase stably transfected into the human follicular thyroid carcinoma cell line FTD-238, was used. Benzophenone-2 (300 nmol/l) combined with the thyroid peroxidase substrate hydrogen peroxide (10 µmol/l) inactivated human recombinant thyroid peroxidase.

In vivo

It has been reported that the derivative 2,2_,4,4_ tetrahydroxy-benzophenone (BP2) exerts considerable estrogenic effects in uterus, vagina, bone and liver of ovariectomised (ovx) rats (Schlecht *et al.*, 2004; Seidlova-Wuttke *et al.*, 2004).

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In the study from Schlecht *et al.* (2004) described above in section 3.3.2: compared to the control group, mean uterine weight was increased significantly in the BP2-100, BP2-333 and BP2-1000 groups. The authors concluded that despite its rapid conjugation and excretion, the levels of the remaining free BP2 in rat serum are sufficient to induce an unequivocal estrogen-like effect in the uterus (Table 1 below).

Table 1
Mean uterus weights of BP2 [mg/kg bodyweight] or vehicle treated rats after five days of treatment

Vehicle	BP2-10	BP2-33	BP2-100	BP2-333	BP2-1000
86.3 ± 3.1	89.5 ± 3.6	87.9 ± 3.3	123.9 ± 6.1*	222.4 ± 9.8*	298.5 ± 8.7*

* $p < 0.05$ versus vehicle (olive oil treated rats), $n = 5$ rats per group.

In a recently published work, orally administered BP2 to pregnant mice was shown to induce hypospadias in the male offspring via signalling through the estrogen receptors (Hsieh *et al.*, 2007).

Co-administration of BP2 and the estrogen-antagonist EM-800 abolished this genital malformation, indicating that also *in vivo* BP2 activates ERs. The authors suggested that the effect of BP2 on uterus weight reported in the study, and on bone morphology, lipid metabolism and gonadotropin secretion shown previously are mediated via binding to ERs (Seidlova-Wuttke *et al.*, 2004).

Further evidence for an estrogenic endocrine disrupting activity of BP2 on secondary sex characteristics, gonadal development and reproduction derived from experiments with fish. After exposure of reproductively mature fathead minnows to 0.002, 0.1, 1.2, 5.0 and 9.7 mg/L BP2 for 15 days, at concentrations of 1.2 mg/L and higher, spermatocyte and oocyte development was significantly inhibited (Weisbrod *et al.*, 2007).

Testes of exposed males had much fewer spermatocytes and ovaries of exposed females contained less mature but more atretic follicles. Reproduction was negatively affected in a dose-dependent manner with a decrease in egg production at 5.0 mg/L and a complete cessation of spawning activity at 9.7 mg/L BP2. Expression of vitellogenin, a characteristic estrogen regulated gene, was also induced by BP2 (Kunz *et al.*, 2006).

The endocrine activation potential of Benzophenone-2 was evaluated using groups of 11 ovariectomized adult Sprague-Dawley rats (Jarry, 2004). The test groups were dosed orally (by gavage) with 250 mg/kg and 1000 mg/kg Benzophenone-2 (in 1 ml olive oil) daily for 5 d. Another group was dosed with E2 valerate (600 µg/kg) according to the same procedure. Control animals were dosed with olive oil. Dosing was initiated 14 d after ovariectomy. Average food intake was significantly reduced during the treatment period. However, there were no differences in liver, spleen, nor adrenal weights between test and control groups. Dosing with E2 valerate resulted in significantly increased uterine weight. Both doses of Benzophenone-2 also had this effect on the uterus. Blood luteinizing hormone levels were statistically significantly reduced after dosing with E2 valerate and 1000 mg/kg Benzophenone-2. There was no evidence of changes in mRNA levels of gonadotropin releasing hormone in the preoptic area of the hypothalamus. A dose-dependent suppression of T4 concentration by Benzophenone-2 was observed. T3 levels were also reduced.

A dose-response experiment involving 5 doses (10, 33, 100, 333, or 1000 mg/kg) of Benzophenone-2 was performed using female Sprague-Dawley adult, ovariectomized rats (groups of 5 (Schlecht *et al.*, 2004). Doses were administered (by gavage) once per day for 5 d. Free levels of Benzophenone-2 in rat serum were sufficient to induce an unequivocal estrogen-like effect in the uterus. When compared to the vehicle (olive oil) control group,

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mean uterine weight was increased statistically significantly in the 333 mg/kg and 1000 mg/kg dose groups.

A similar study (groups of 12; same doses and protocol) involved ovariectomized rats of the same strain. E2 valerate served as the positive control. None of the animals showed clinical signs of toxicity. Benzophenone-2 exerted an estrogenic effect on the following uterine parameters at the administered doses: wet weight, complement protein 3 (C3), insulin-like growth factor (IGF1), and estrogen receptor β (Er β) gene expression. (Schlecht *et al.* 2006,)

Benzophenone-2 was evaluated for its effect on the hypothalamic-pituitary-thyroid (HPT) axis (Broniowska *et al.*, 2018). The test substance was dissolved in a small amount (volume not stated) of ethanol and olive oil and formulated with Hascobase. The test substance was then applied to shaved skin of 10 male Wistar rats at a dose of 100 mg/kg for 4 wk. HPT activity was increased, *i.e.*, the level of thyroid-stimulating hormone (TSH) was reduced and the free fraction of T3 and T4 in the blood was increased.

Benzophenone-2 interference with thyroid function was also studied *in vivo* (Schmutzler *et al.*, 2007). Groups of 12 adult female Sprague-Dawley rats were bilaterally ovariectomized and fed a soy-free diet containing iodide ad libitum. At 14 d after ovariectomy, groups of 12 rats were dosed orally (by gavage, once per day) with Benzophenone-2 at the following doses (dose volume = 1 ml): 10 mg/kg, 33 mg/kg, 100 mg/kg, 333 mg/kg, and 1000 mg/kg. The animals were killed at day 5, and thyroid glands were excised. A dose-dependent decrease in total serum T4 levels was observed, with statistically significant alterations at doses of 333 mg/kg and 1000 mg/kg. The small decrease in total T3 was not statistically significant. TSH levels were increased at doses of 333 mg/kg and 1000 mg/kg, and this increase was statistically significant at both doses. Thyroid peroxidase activities in the thyroid glands of treated animals were measured *ex vivo*, but no statistically significant dose-dependent changes were observed. In the livers of animals treated with 1000 mg/kg Benzophenone-2, type I 5'-deiodinase activity was decreased, and this decrease was statistically significant. However, an increase in type I 5'-deiodinase activity was observed at a dose of 33 mg/kg.

Overall SCCS comment on ED effects of Benzophenone-2

The available evidence from data in various species shows that BP-2 acts as an estrogenic compound and can alter thyroid hormone balance by affecting the central regulation and metabolism of the hormones.

Effect on Melanogenesis - Benzophenone-2

The dual action of Benzophenone-2 in the biosynthetic pathway of melanin has been identified (Garcia-Jimenez *et al.*, 2017). It has been observed to act as a weak competitive inhibitor of tyrosinase (inhibition constant (K_i) = $2020 \pm 90 \mu\text{M}$; half maximal inhibitory concentration (IC_{50}) = $3820 \pm 390 \mu\text{M}$). Both forms of Benzophenone-2 (protonated and deprotonated) interact with tyrosinase, the enzyme that catalyzes the production of melanin from tyrosine. Benzophenone-2 (at 250 and 500 μM) also accelerated the conversion of dopachrome (intermediate in melanin biosynthesis) to melanin.

Neurotoxicity- Benzophenone-2

A study was performed, using groups of 10 male Wistar rats, to determine apoptosis and oxidative stress markers in the rat brain after topical administration of Benzophenone-2 (Broniowska *et al.*, 2019). The markers studied were: active form of caspase-3, pro-apoptotic protein (Bax), and anti-apoptotic protein (Bcl-2). The effect of dosing on these markers was studied to determine whether Benzophenone-2 may be involved in the induction or exacerbation of neurodegenerative changes. Benzophenone-2 was dissolved in a small amount (volume not stated) of ethanol and olive oil, and formulated with Hascobase. The test

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substance was then applied to shaved skin at a dose of 100 mg/kg for 4 wk. Hascobase with a small amount of ethanol and olive oil was applied to the skin of control rats. In the hippocampus, where the Benzophenone-2 concentration was ~3.5-fold lower than in the frontal cortex, no statistically significant changes in oxidative stress and apoptosis markers were observed. In the frontal cortex, there was no change in apoptosis markers, but, unexpectedly, the oxidative stress markers were reduced. The authors concluded that Benzophenone-2 did not exacerbate oxidative stress and apoptosis markers in the hippocampus and frontal cortex. However, it did lower oxidative stress in the frontal cortex.

The effect of Benzophenone-2 and Benzophenone-3 on the neuroblastoma (SH-SY5Y) cell line was evaluated by studying effects on cell viability and caspase-3 (main executive enzyme in programmed cell death) activity.⁹⁷ The MTT reduction test and LDH release activity assay were used. After a 72-h incubation period, both Benzophenone-2 and Benzophenone-3 produced a statistically significant cytotoxic effect at concentrations of 10 µM and 100 µM in both assays. Additionally, both Benzophenone-2 and Benzophenone-3 caused an increase in caspase-3 activity at much lower concentrations (from 0.01 µM to 0.1 µM). The authors noted that the results of this study indicate that Benzophenone-2 and Benzophenone-3 adversely affected the viability of nerve cells, most likely by enhancing the process of apoptosis

Immunomodulatory Effects - Benzophenone-2

The *in vitro* effect of Benzophenone-2 on the production of interferon (IFN)- γ and interleukin (IL)-10 was studied (Rachon *et al.*, 2006). IFN- γ and IL-10 are two cytokines representing the Th-1 lymphocyte and Th-2 lymphocyte response, respectively, by activated murine splenocytes. Splenocytes were cultured in the presence of different concentrations of Benzophenone-2 (0.01 to 10 µM). Benzophenone-2 (10 µM) shifted the Th1/Th2 balance toward a Th2 response (lower IFN- γ production and higher IL-10).

Benzophenone-2 (in ethanol) was administered dermally (100 mg/kg), twice daily for 4 wk, to 10 male Wistar rats (Broniowska *et al.*, 2018). Immunological parameters were assayed 24 h after the last administration. Dosing with Benzophenone-2 did not change relative weights of the spleen and thymus, and was not toxic to splenocytes and thymocytes. However, dosing did increase the proliferative activity of splenocytes, and also enhanced the metabolic activity and viability of splenocytes and thymocytes.

3.6 TOXICOLOGICAL EVALUATION (BP-5)

As explained in section 3.2, the SCCS used the toxicological data from BP-4 for read-across to BP-5. The summaries drawn from conclusions of the SCCS Opinion on BP-4 (SCCS/1660/23) are given below for each toxicological endpoint. By means of read-across, these are also considered applicable to BP-5 by the SCCS.

3.6.1 Skin irritation and corrosivity

According to the existing *in vitro* information, neat benzophenone-4 is corrosive for the skin. However, when diluted in water, benzophenone-4 is non-irritating to rabbits at a concentration of 5% in and up to 10% in humans. This conclusion was reached earlier by the SCCNFP.

3.6.2 Eye irritation and corrosivity

An *in vitro* and *in vivo* eye irritation/ corrosion study were available. Although not carried out according to the Notes of Guidance, they both pointed to eye irritative properties of benzophenone-4, as also indicated earlier by ECHA and to be expected by the acidic pH.

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3.6.3 Skin sensitisation

Although animal studies show skin sensitisation potential to BP-4, the risk of sensitisation to humans can be regarded as very low.

3.6.4 Acute toxicity

The studies referred to in the SCCS Opinion (SCCS/1660/23) pointed to low or no acute toxicity of benzophenone-4 in animals. However, as indicated in the SCCS Notes of Guidance, acute toxicity studies are not mandatory for risk assessment of cosmetic ingredients.

3.6.5 Repeated dose toxicity

From the 90-day repeat dose oral toxicity study in rats, referred to in the SCCS Opinion (SCCS/1660/23) a NOAEL of 1239 mg/kg bw/day could be derived for BP-4.

3.6.6 Reproductive toxicity

The OECD TG 422 study, referred to in the SCCS Opinion (SCCS/1660/23) for BP-4, showed no treatment related changes up to 1250 mg/kg bw /day.

3.6.7 Mutagenicity / genotoxicity

Benzophenone-4 has no *in vitro* genotoxic potential.

3.6.8 Carcinogenicity

No data available.

3.6.9 Photo-induced toxicity

Benzophenone-4 is not phototoxic, not photosensitising, not photomutagenic and not photoclastogenic.

3.6.10 Human data

Considering the available information in humans, the risk of sensitisation is very low.

3.6.11 Special investigation

No indications of endocrine activity.

3.7 SAFETY EVALUATION (INCLUDING CALCULATION OF THE MOS)

Benzophenone-2

Having considered all the available data, the SCCS is of the view that assessment of the safety of BP-2 cannot be carried out because the genotoxicity potential cannot be excluded.

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Benzophenone-5

As explained in section 3.2, the SCCS considers it appropriate to use the available toxicological data on BP-4 for read-across to BP-5. The calculation of the systemic exposure dose (SED), the margin of safety (MoS), and the selection of Point of Departure (PoD) for BP-5 are therefore taken as the same for BP-4, which has already been assessed in the SCCS Opinion (SCCS/1660/23).

Table 7: Margins of Safety (MoS) calculation for benzophenone-5 based on read-across from BP-4 when used in cosmetic products at 5%.

Product type	SED (mg/kg bw/day)	Adjusted NOAEL (mg/kg bw/day)	MoS
Dermal exposure:			
Sunscreen lotion	0.069	620	8986
Face cream*	0.006	620	103333
Hand cream*	0.008	620	77500
Oral exposure:			
Lip stick*	0.045	620	13778
Inhalatory exposure:			
Sunscreen propellant spray*	0.088	620	7046
Sunscreen pump spray	0.001	620	620000
Overall aggregate* (deterministic)	0.147	620	4218

*Aggregate exposure has been calculated from face cream, hand cream, lipstick and sunscreen propellant spray (only one sunscreen was considered for the worst-case scenario)

The calculation has shown that the Margin of Safety (MoS) is above 100 for the use of benzophenone-5 at a maximum concentration of 5% in different cosmetic products that are applied directly on the skin, or in sunscreen formulations in propellant or pump spray forms, whether these products are used individually or in combination.

3.8 DISCUSSION

As explained in section 3.2, due to the scarcity of available information on BP-5, the SCCS has considered it appropriate to use data from a close analogue BP-4 for read-across for use in the safety assessment of BP-5.

Physicochemical properties

The physical form of Benzophenone-2 has been described as yellow needle-like crystals, and Benzophenone-5 as white or light-yellow powder. Only limited data on the heavy metal impurities of Benzophenone-2 are available. Full reports on the purity and impurities in representative batches of both Benzophenone-2 and Benzophenone-5 on the market are not available.

The available information, including model estimated values, indicate that Benzophenone-2 is slightly soluble in water, whereas Benzophenone-5 is freely soluble in water. The partition coefficient (Log Pow) of Benzophenone-2 has been estimated at 2.64 (predicted average) and -1.26 for Benzophenone-5.

Data on the stability under use conditions in cosmetic product are not available for both benzophenone-2 and benzophenone-5. However, both BP-2 and BP-5 have been advised to be stored in a closed system and kept in a dark place without any light exposure.

Toxicokinetics

The available data indicate that a large proportion of the applied Benzophenone-2 will be rapidly metabolised and excreted. Although no data/information relating to ADME of Benzophenone-5 are available, considering the high solubility in water, any systemically absorbed Benzophenone-5 can be expected to be rapidly cleared and excreted via urine.

Exposure

Benzophenone 2:

Only old (1983) data are available on the use, and the levels of use, of Benzophenone-2 in different cosmetic products. In the absence of more recent data, the SCCS has used the available information in Table 4 and considered the use of Benzophenone-2 in all cosmetic products at 1% maximum.

Benzophenone 5:

Data on the use of Benzophenone-5 in different cosmetic products, and the concentrations used, are not available. However, since Benzophenone-5 is allowed in ready for use cosmetic preparations at concentrations up to 5% (as acid) (Annex VI/22 of the cosmetic regulation), the SCCS has used this concentration for the calculation of Systemic Exposure Dose (SED).

Toxicological Evaluation

Irritation and corrosivity

Benzophenone 2:

Benzophenone-2 has been reported to be slightly to moderately irritating at 100% concentration, but non-irritating when tested at 16% in dimethyl phthalate (DMP) or petrolatum.

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Benzophenone 5:

Eye irritative properties of benzophenone-4 (the analogue used for read-across to BP-5) have been indicated, as expected from the presence of the sulfonic acid moiety. BP-4 has also been reported to be corrosive when applied neat to the skin, but not when diluted in water.

Skin sensitisation

Benzophenone 2:

In view of only a limited number of reported cases in the literature compared to the potential exposure of a large number of individuals, the risk of sensitisation to Benzophenone-2 can be regarded as low.

Benzophenone 5:

Data from BP-4 (the analogue used for read-across to BP-5) indicate that although animal studies show skin sensitisation potential, the risk of sensitisation can be regarded as low for humans.

Acute toxicity

Acute toxicity studies are not mandatory for risk assessment of cosmetic ingredients. However, the data from available studies have indicated a low or no acute toxicity in animals of both Benzophenone-2 and Benzophenone-4 (the analogue used for read-across to BP-5).

Repeated dose toxicity

Only limited data on repeated-dose toxicity of Benzophenone-2 are available. For BP-4 (the analogue used for read-across to BP-5), a NOAEL of 1239 mg/kg bw/day could be derived from a 90-day repeat dose oral toxicity study in rats.

Reproductive toxicity

Benzophenone 2:

The information on reproductive toxicity of Benzophenone-2 is limited to one study that reported hypospadias in male fetuses of the treated pregnant female mice and hypothesised this to be via signalling through the estrogen receptor. The authors, however, highlighted the need for further study at human benzophenone-2 exposure and its effects to support this hypothesis.

Benzophenone 5:

For BP-4 (the analogue used for read-across to BP-5), no treatment related changes were noted up to 1250 mg/kg bw/day.

Mutagenicity / genotoxicity

Benzophenone 2

Most of the data on genotoxicity/mutagenicity testing of BP-2 analysed by the SCCS were derived from study reports to which the SCCS had no access. Although, one study on *in vitro* SCE reported positive results, an *in vivo* SCE study reported negative results. Other genotoxicity endpoints such as *in vitro* Comet assay, or *in vitro* UDS have indicated negative results. Considering all the available information, the SCCS has concluded that genotoxicity/mutagenicity of BP-2 potential cannot be excluded due to:

- positive results reported in gene mutations tests in different bacterial strains (Ames test),
- positive results reported in gene mutations in mammalian cells (Mouse Lymphoma Assay),

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- positive results reported in one chromosomal aberration test (mouse lymphoma L5178Y cells).

Benzophenone 5:

The available data on BP-4 (the analogue used for read-across to BP-5) has indicated that it does not have *in vitro* genotoxic potential.

Carcinogenicity

No data on carcinogenicity are available for either Benzophenone-2 or Benzophenone-5.

Photo-induced toxicity

Benzophenone 2:

In view of the limited number of reported cases in the literature from the potential exposure of many individuals, the SCCS considers that the risk of photo-induced toxicity to Benzophenone-2 can be regarded as low.

Benzophenone 5:

Benzophenone-4 (the analogue used in read-across to BP-5) is not phototoxic, not photosensitising, not photomutagenic and not photoclastogenic.

Human data

Considering the available information in humans, the risk of sensitisation to either BP-2 or BP-5 is regarded as low.

Special investigation

Benzophenone 2:

The available data in various species indicate that BP-2 acts as an estrogenic compound and can alter thyroid hormone balance by affecting the central regulation and metabolism of the hormones.

Benzophenone 5:

The available data on BP-4 (the analogue used for read-across to BP-5) has not indicated endocrine activity.

4. CONCLUSION

In light of technical and scientific progress, available scientific literature, relevant in silico tools and the SCCS' expert judgement and taking under consideration in particular the concerns related to potential endocrine disrupting properties, the SCCS is requested:

- 1. to identify and justify whether there are specific concerns regarding the use of Benzophenone-2 and Benzophenone-5 in cosmetic products.*

Benzophenone 2 (BP-2)

Having considered all the available data, and the concerns relating to potential endocrine disrupting properties, the SCCS cannot conclude on the safety of Benzophenone-2 because genotoxicity potential cannot be excluded. The data on repeated dose toxicity and reproductive toxicity of Benzophenone-2 are also either limited or not available. The available evidence shows that BP-2 is an endocrine-active substance, due to a clear demonstration of estrogenic activity both *in vitro* and *in vivo*.

Benzophenone 5 (BP-5)

Due to the scarcity of information for Benzophenone-5, the SCCS has considered that data from a close analogue BP-4 can be used for read-across for the safety assessment of BP-5. Thus, considering the toxicological data on Benzophenone-4, including the potential endocrine disrupting properties, the SCCS is of the opinion that BP-5 is safe when used as a UV filter up to a maximum concentration of 5% in sunscreen, face and hand cream, lipstick, sunscreen propellant spray and pump spray, when used separately or in combination (based on deterministic aggregated exposure of BP-4).

- 2. to highlight if there is a potential risk for human health from the use of Benzophenone-2 and Benzophenone-5 in cosmetic products.*

The SCCS considers that the use of Benzophenone-2 in cosmetic products may pose a risk to the consumer because of the indications for genotoxicity and endocrine activity. The assessment of safety of BP-2 in cosmetic products also requires appropriate data on repeated dose toxicity and reproductive toxicity.

The SCCS mandate does not address environmental aspects. Therefore, this assessment has not covered the safety of BP-2 and BP-5 for the environment.

5. MINORITY OPINION

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7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

Scientific advice on the safety of Benzophenone-2 (BP-2) (CAS No. 131-55-5, EC No. 205-028-9) and Benzophenone-5 (BP-5)(CAS No. 6628-37-1, EC No. 613-918-7) as substances with potential endocrine disrupting properties in cosmetic products

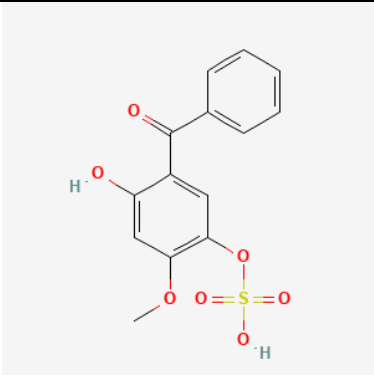
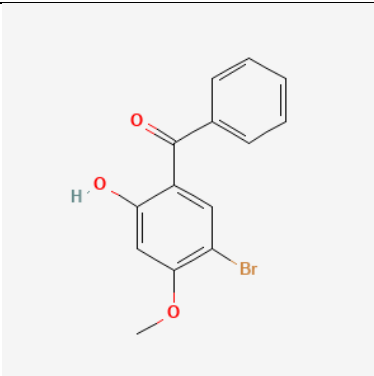
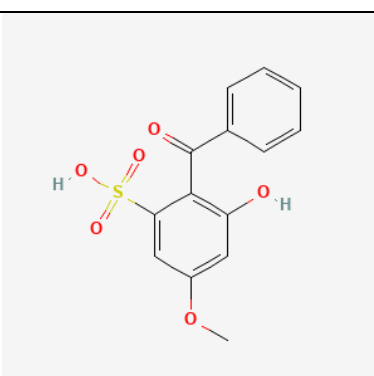
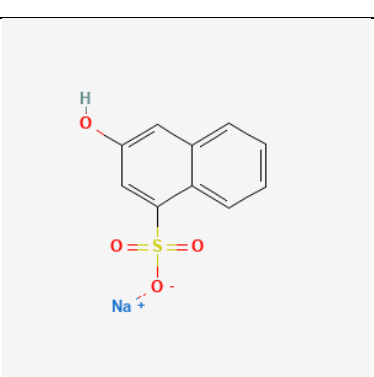
ANNEX BP-2 and BP-5

Table A:

Identification of analogues for read-across assessment of benzophenone-5 (BP-5)

CAS number	SMILES	Name	Similarity (%)	Chemical Structure
6628-37-1	<chem>[Na+].COc1cc(O)c(cc1S([O-]))(=O)=O)C(=O)c1ccccc1</chem>	Benzophenone-5 (The target substance)	100	
4065-45-6	<chem>COc1cc(O)c(cc1S(O)(=O)=O)C(=O)c1ccccc1</chem>	Benzophenone-4 (Sulisobenzene)	88.3	
No specific CAS (same CAS used as of BP-4)	<chem>O.COc1cc(O)c(cc1S(O)(=O)=O)C(=O)c1ccccc1</chem>	Sulisobenzene hydrate (hydrated form of BP-4)	86.36	

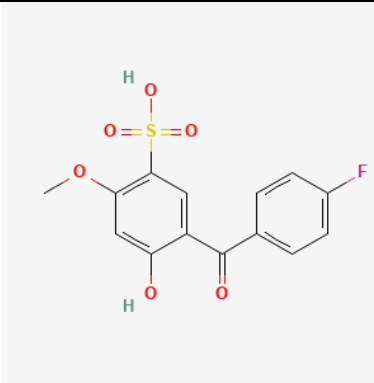
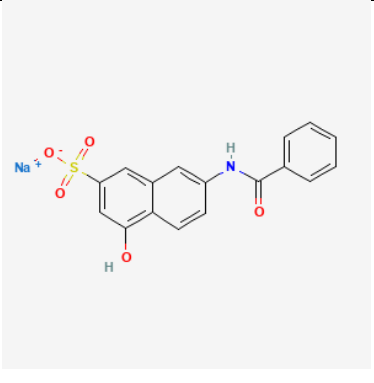
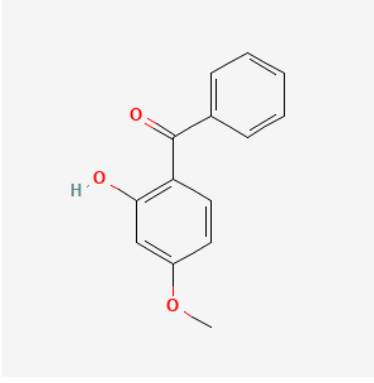
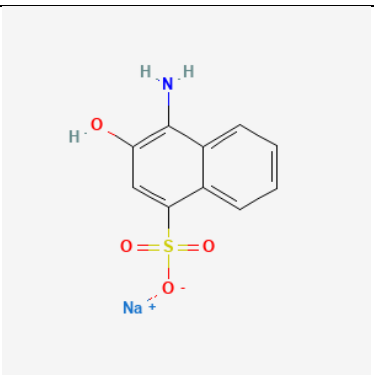
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84055-67-4	<chem>COc1cc(O)c(cc1OS(=O)(=O)C(=O)c1ccccc1</chem>	Methanone, (2-hydroxy-4-methoxy-5-(sulfooxy)phenyl)phenyl-	81.82	
3286-93-9	<chem>COc1cc(O)c(cc1Br)C(=O)c1ccccc1</chem>	5-Bromo-2-hydroxy-4-methoxybenzophenone	80.0	
112501-13-0	<chem>COc1cc(O)c(C(=O)c2ccc(c2)c1S(=O)(=O)O</chem>	2-Benzoyl-3-hydroxy-5-methoxybenzenesulfonic acid	79.07	
93804-66-1	<chem>[Na+].[O-]c1cc2ccccc2c(c1)S(=O)(=O)[O-]</chem>	Sodium 3-hydroxynaphthalene sulphonate	78.95	

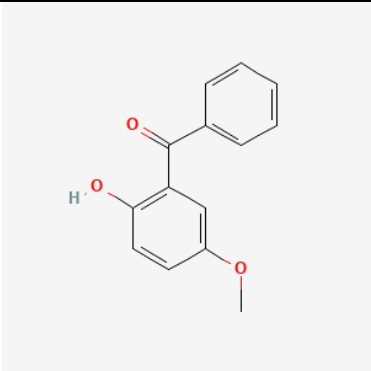
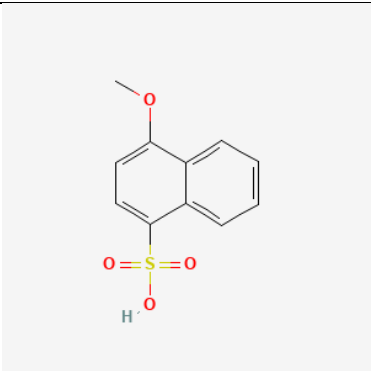
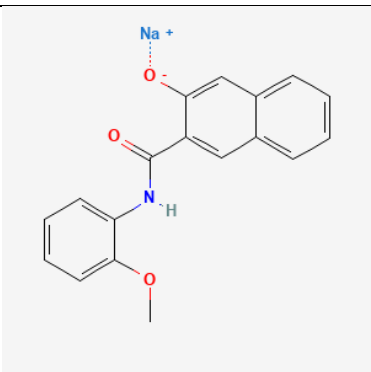
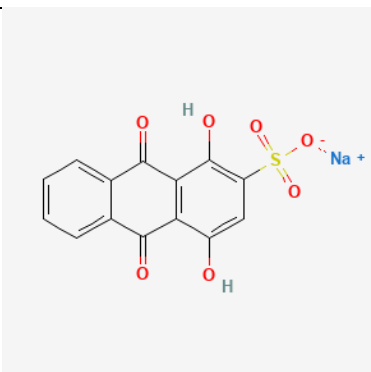
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13935-00-7	<chem>[Na+].Oc1cc(cc2ccccc12)S([O-])(=O)=O</chem>	sodium 4-hydroxynaphthalene-2-sulfonate	4	78.95	
3121-60-6	<chem>[Na+].COc1ccc(c(O)c1)C(=O)c1cc(c(OC)cc1O)S([O-])(=O)=O</chem>	2,2'-dihydroxy-4,4'-dimethoxy-5-sulfo benzophenone sodium salt		76.60	
84145-65-3	<chem>[Na+].[Na+].Oc1cc2c(cccc2S([O-])(=O)=O)cc1C([O-])=O</chem>	Disodium hydroxy-5-sulphonato-2-naphthoate	3-	76.19	
59687-31-9	<chem>[Na+].Oc1ccc(OC(=O)c2ccccc2)cc1S([O-])(=O)=O</chem>	5-(Benzoyloxy)-2-hydroxybenzenesulfonic acid sodium salt		74.42	

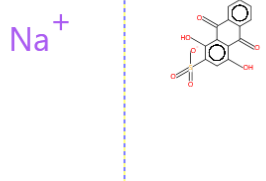
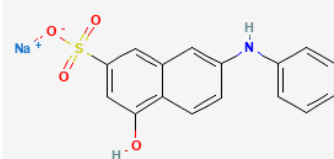
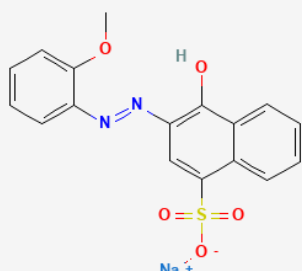
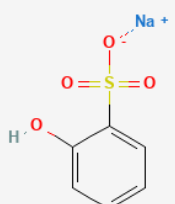
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84794-98-9	<chem>COc1cc(O)c(cc1S(=O)(=O)C(=O)c1ccc(F)cc1</chem>	5-(4-fluorobenzoyl)-4-hydroxy-2-methoxybenzenesulfonic acid	72.73	
10534-92-6	<chem>[Na+].Oc1cc(cc2cc(NC(=O)c3ccccc3)ccc12)S([O-])(=O)=O</chem>	2-Naphthalenesulfonic acid, 7-(benzoylamino)-4-hydroxy-, monosodium salt	72.34	
131-57-7	<chem>COc1ccc(C(=O)c2ccccc2)c(O)c1</chem>	Benzophenone-3 (Oxybenzone)	71.79	
5959-58-0	<chem>[Na+].Nc1c(O)cc(c2ccccc12)S([O-])(=O)=O</chem>	1-Amino-2-naphthol-4-sodium sulfonate	71.79	

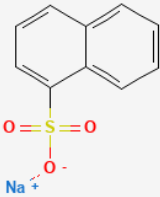
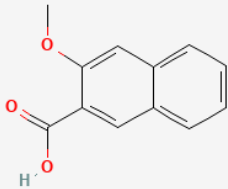
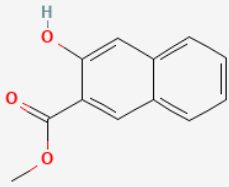
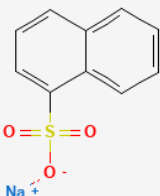
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14770-96-8	<chem>COc1ccc(O)c(c1)C(=O)c1ccccc1</chem>	Benzophenone, 2'-hydroxy-5'-methoxy-	71.79	
84473-60-9	<chem>[Na+].COc1ccc(c2ccccc12)S([O-])(=O)=O</chem>	sodium 4-methoxynaphthalene-1-sulfonate	71.79	
68556-01-4	<chem>[Na+].COc1cccc1NC(=O)c1cc2ccccc2cc1O</chem>	sodium N-(o-anisyl)-3-hydroxynaphthalene-2-carboxamidate	71.11	
22297-70-7	<chem>[Na+].Oc1cc(c(O)c2C(=O)c3ccccc3C(=O)c12)S([O-])(=O)=O</chem>	Sodium 2-quinizarinsulfonate	71.11	

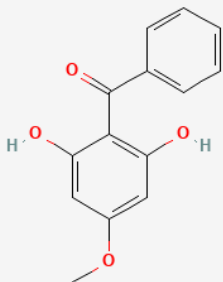
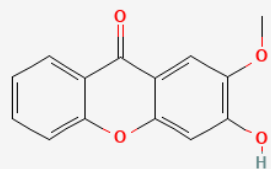
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33796-05-3	<chem>[Na+].Oc1cc(c(O)c2C(=O)c3ccccc3C(=O)c12)S([O-])(=O)=O</chem>	1,4-dihydroxy-2-sulfoanthraquinone sodium salt	71.11	
68213-89-8	<chem>[Na+].Oc1cc(cc2cc(Nc3ccccc3)ccc12)S([O-])(=O)=O</chem>	2-Naphthalenesulfonic acid, 4-hydroxy-7-(phenylamino)-, monosodium salt	71.11	
5858-39-9	<chem>[Na+].COc1ccccc1N=Nc1cc(c2ccccc2c1O)S([O-])(=O)=O</chem>	C.I. Acid Red 4, monosodium salt	70.83	
1300-51-2	<chem>[Na+].Oc1ccccc1S([O-])(=O)=O</chem>	Benzenesulfonic acid, hydroxy-, monosodium salt	70.59	

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130-14-3	<chem>[Na+].[O-]S(=O)(=O)c1cccc2ccccc21</chem>	1-Naphthalenesulfonic acid, sodium salt	70.27	
883-62-5	<chem>COC1C=CC(=O)C2=CC=CC=C2C1=O</chem>	3-methoxy-2-naphthoic acid	70.27	
883-99-8	<chem>COC(=O)c1cc2ccccc2cc1O</chem>	methyl 3-hydroxy-2-naphthoate	70.27	
1321-69-3	<chem>[Na+].[O-]S(=O)(=O)c1cccc2ccccc21</chem>	Naphthalenesulfonic acid, sodium salt	70.27	

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479-21-0	<chem>COc1cc(O)c(C(=O)c2ccccc2)c(O)c1</chem>	Cotoin (2,6-Dihydroxy-4-methoxybenzophenone)	70.00	
33018-28-9	<chem>COc1cc2C(=O)c3ccccc3Oc2cc1O</chem>	3-Hydroxy-2-methoxyxanthen-9-one	70.00	
18799-43-4	<chem>COc1cc(O)c2C(=O)c3ccccc3Oc2c1</chem>	Xanthen-9-one, 1-hydroxy-3-methoxy-	70.00	